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RESEARCH ARTICLE
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The Late Miocene Ocean Carbon Cycle Changes: New Evidence From the South Atlantic IODP Expeditions

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Key Points:

- This study presents new late Miocene benthic foraminiferal $\delta^{13}C$ and $\delta^{18}O$ in the South Atlantic
- There is an intensified basinal gradient in the deep ocean $\delta^{13}C$ over the late Miocene
- The intensified basinal $\delta^{13}C$ gradient can result from sluggish or weakening of ocean overturning circulation

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The late Miocene ocean carbon cycle changes can be characterized by the secular 405-kyr long-eccentricity cycles in ocean dissolved inorganic carbon isotopic composition ($\delta^{13}C$), superimposed by a negative shift in ocean $\delta^{13}C$, namely, the Late Miocene Carbon Isotope Shift. During this epoch, the Earth experienced a long-term cooling, especially in high-latitude oceans. Comparatively, atmospheric CO_2 reconstructions do not show a paralleled decline as the late Miocene cooling, suggesting a decoupling of CO_2 radiative forcing and Earth's climate. Analysis on ocean carbon cycle and circulation changes can provide insights into the late Miocene cooling. Here we present new benthic foraminiferal isotopes from International Ocean Discovery Program Site U1560 in Hole A (U1560A) in the South Atlantic. Together with existing benthic foraminiferal $\delta^{13}C$ records, we examine the pattern of global deep ocean $\delta^{13}C$ over the late Miocene. Our analysis shows a step-wise increase in the benthic $\delta^{13}C$ difference between the Atlantic and Indian-Pacific Oceans. The increased basinal deep ocean $\delta^{13}C$ gradient agrees with a positive shift in the benthic $\delta^{18}O$ in Hole U1560A, which may indicate deep ocean cooling or enhanced Antarctic glaciation. As supported by lines of geological evidence and a diagnostic carbon cycle box-model developed in this study, sluggish or weakening of ocean meridional overturning circulation due to enhanced Antarctic glaciation can be a driver for the increased $\delta^{13}C$ gradient. Such a change in ocean circulation may reduce the polarward ocean heat transport, leading to deepened mixed layer depth in the tropical oceans.

Plain Language Summary The oxygen and carbon isotopes ($\delta^{18}O$ and $\delta^{13}C$) of benthic foraminifer can record the isotopic signatures of seawater and dissolved inorganic carbon in deep ocean. A higher benthic $\delta^{18}O$ indicates cooler deep water and/or that more ice with lower $\delta^{18}O$ is trapped in land. Benthic $\delta^{13}C$ can reflect the transfer or cycling of carbon between the biosphere, hydrosphere, atmosphere, and lithosphere reservoirs. In this study, we present new benthic foraminifer $\delta^{18}O$ and $\delta^{13}C$ records from the marine sediments which are deposited during the late Miocene period, approximately 5–10 million years ago, at an ocean drilling site in the South Atlantic. By comparing our data with other published late Miocene benthic $\delta^{13}C$, we find a generally increased deep ocean $\delta^{13}C$ gradient between the Atlantic and Pacific Oceans over the late Miocene. This phenomenon can reveal a reorganization of global deep ocean circulation.

1. Introduction

The late Miocene (5.33–11.65 million years ago, Ma) ocean carbon cycle changes can be decomposed into two aspects (a) cyclic changes in seawater dissolved inorganic carbon isotopic composition ($\delta^{13}C$) corresponding to the 405-kyr long-eccentricity cycles; (b) and a negative shift in ocean $\delta^{13}C$ in a magnitude of ~0.8‰ (Westerhold et al., 2020) (Figure 1). The latter is referred to as the Late Miocene Carbon Isotope Shift (LMCIS), which initiated from 7.6 Ma and lasted for approximately 1 million years (Myr) in the Pacific Ocean (Drury et al., 2017). Climate changes during the late Miocene are characterized by a long-term decline in mid- and high-latitude ocean and land surface temperatures (Herbert et al., 2016; Wen et al., 2023), suggesting a late Miocene cooling (Figure 1). In comparison, the globally composited benthic foraminiferal oxygen isotopes ($\delta^{18}O$) do not show a parallel increase which is supposed to indicate cooling of deep water and/or greater land ice-sheet covering (Westerhold et al., 2020; Zachos et al., 2001) (Figure 1). Some studies also predict that there was supposed to be a decline in atmospheric CO_2 concentration over the late Miocene to drive the long-term cooling (Wen et al., 2023)

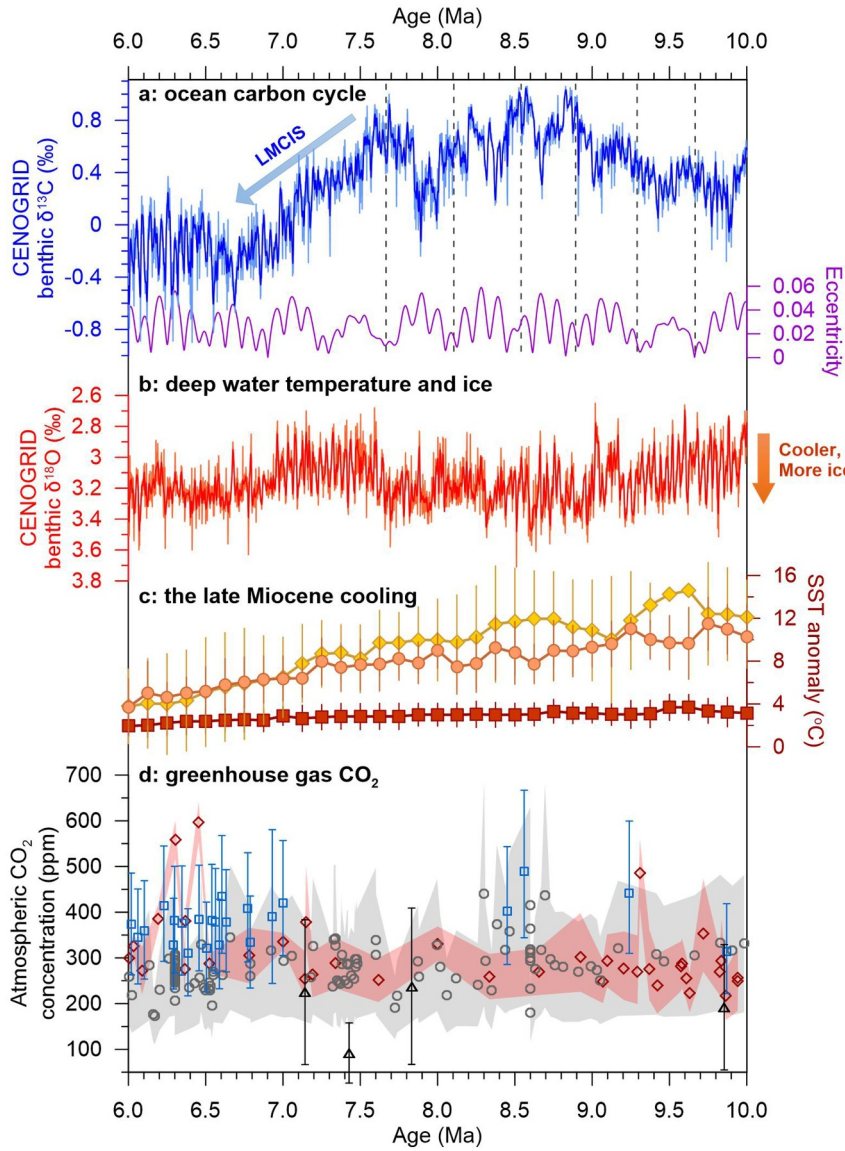


Figure 1. The late Miocene ocean carbon cycle and climate changes: (a) the Cenozoic Global Reference benthic foraminifer carbon and oxygen Isotope Data set (CENOGRID) $\delta^{13}C$ (Westerhold et al., 2020) plotted versus eccentricity (Laskar et al., 2004) showing (i) the secular 405-kyr long-eccentricity carbon cycles (dashed lines), and (ii) the Late Miocene Carbon Isotope Shift (LMCIS); (b) The CENOGRID $\delta^{18}O$: higher (lower) $\delta^{18}O$ indicating cooler (warmer) deep ocean temperature and/or more (less) land ice covering; (c) The Herbert et al. (2016) sea surface temperature (SST) anomaly stacks of tropics (□), high-latitude oceans in the north (◇) and south (○) hemispheres, with error bars indicating one standard deviation; (d) Atmospheric CO_2 reconstructions from planktonic foraminiferal boron (□, $\pm 2\sigma$) (Brown et al., 2022; Sosdian et al., 2018), alkenones (◇) with the 95% uncertainty shading (red) (Bolton et al., 2016; Super et al., 2018; Y. G. Zhang et al., 2017), land plants (◇) with the 95% uncertainty shading (gray) (Cui et al., 2020), and paleosol (Δ, $\pm 1\sigma$) (Ji et al., 2018). The late Miocene CENOGRID are obtained by merging the records of IODP Site U1337 (Drury et al., 2017; Tian et al., 2018) in the equatorial Eastern Pacific and ODP Site 1146 (A. E. Holbourn et al., 2018) in the South China Sea in the tropical Western Pacific.

and the associated atmospheric circulation changes, for example, intensified East Asian Winter Monsoon (Q. Zhang et al., 2024), or to be a consequence of the LMCIS (Du et al., 2022). However, our current knowledge is that the atmospheric CO_2 concentration could have a range of ~250–450 ppm (CenCO2PIP-Consortium, 2023), showing no paralleled decline with the late Miocene cooling and the LMCIS (Figure 1). In fact, the CO_2 reconstructions show incompatible results from various proxies, for example, planktonic foraminiferal boron isotopes (Brown et al., 2022; Sosdian et al., 2018), marine alkenones (Bolton et al., 2016; Super et al., 2018;

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Tanner et al., 2020; Y. G. Zhang et al., 2017), paleosol (Ji et al., 2018), and land plants (Cui et al., 2020). It can be inferred that the Earth surface temperature decoupled with atmospheric CO_2 concentration over the late Miocene, which suggests non-CO_2 radiative forcing operated in the climate system. Changes in ocean circulation can be critical for modulating the Earth surface temperature by redistributing global ocean heat content (Winton et al., 2013). Indeed, the roles of ocean circulation change in the late Miocene cooling remains unclear. This study delves into ocean carbon cycle and the associated circulation changes, which can provide insights into the late Miocene climate changes.

Ocean carbon cycle changes can be tracked by the epifauna benthic foraminiferal carbon isotopes ($\delta^{13}C$) inherited from dissolved inorganic carbon in deep ocean. The deep ocean $\delta^{13}C$ primarily reveals (a) carbon cycling between the oceanic and other reservoirs, and (b) carbon redistribution within the oceanic reservoir due to the changes in ocean circulation which regulate the transport, mixing, and residence time of deep water in global ocean basins (Cramer et al., 2009). For example, the modern deep ocean $\delta^{13}C$ pattern is governed by the global ocean overturning circulation, namely, the ocean conveyor belt (Talley, 2013) (Figure 2). Formation of deep waters results from the sinking of cold and high-density surface waters in high latitudes, that is, the formation of North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW) (Figure 2). In the Atlantic sector, NADW flows southward and upwells to the surface Southern Ocean. AABW flows northward and mixes with NADW. The newly formed NADW has a positive $\delta^{13}C$ of ~1‰, whereas, the $\delta^{13}C$ of AABW is relatively negative (~0.5‰) (Lauvset et al., 2023) (Figure 2). In the Pacific and Indian sectors, AABW flows northward to further northern latitudes, with even more negative $\delta^{13}C$ (Figure 2). The deep waters then upwell to an intermediate depth, and flow southward to the surface Southern Ocean as an overturning circulation. Finally, the upwelled waters return to the surface Atlantic, forming the global circulation closure (Talley, 2013). Such a pattern of global deep $\delta^{13}C$ mirrors the macronutrient concentration. High seawater nitrate and phosphate contents are coupled with negative $\delta^{13}C$ (Lauvset et al., 2023), because, the light carbon (^{12}C) and nutrients are accumulated in the deep waters due to organic carbon remineralization along the ocean conveyor belt. Therefore, an inter-basin comparison of deep ocean $\delta^{13}C$ can provide hints for ocean carbon cycle and the associated circulation changes.

During the late Miocene, the general configuration of global ocean circulation is similar to the modern (A. Holbourn et al., 2013, 2024). The subsidence of the Greenland-Scotland Ridge since the early Miocene allows the cold Arctic water overflow to the deep Atlantic, as the proto-NADW or the northern-sourced deep water (Wright & Miller, 1996). In the South Hemisphere, enhanced Antarctic glaciation since the Middle Miocene Climate Transition can facilitate the formation of AABW or the southern-sourced deep water (A. Holbourn et al., 2024). Previous studies have shown a greater negative shift in benthic $\delta^{13}C$ in the Southern Ocean and Pacific than Atlantic over the LMCIS (Billups, 2002; Hodell & Venz-Curtis, 2006). The increased basinal deep ocean $\delta^{13}C$ gradient may result from changes in regional ocean productivity, or circulation, for example, a greater contribution of the northern-sourced deep water to the Atlantic basin (Hodell & Venz-Curtis, 2006). Indeed, some new high-resolution benthic $\delta^{13}C$ records have been published at Ocean Drilling Program (ODP) and International Ocean Discovery Program (IODP) sites in the global ocean basins over the past decade (Table 1). Here we present new benthic foraminiferal isotopes ($\delta^{18}O$ and $\delta^{13}C$) at IODP Site U1560 in Hole A (Hole U1560A), retrieved from Expeditions 390 & 393: "South Atlantic Transect." Hole U1560A is located in the central South Atlantic, west of the Mid-Atlantic Ridge (30°24.2064S, 16°55.3718′W) with a modern water depth of 3723.67 m (Coggon et al., 2024) (Figure 2). Since Hole U1560A is also situated in the South Atlantic gyre with oligotrophic and low productivity waters, the benthic foraminiferal $\delta^{13}C$ should primarily indicate a basin-wide change in deep ocean carbon cycle, and to a lesser degree by regional biogeochemical processes in the water column. Based on the benthic isotope records, this study examined the long-term evolution of the global deep $\delta^{13}C$ pattern over the late Miocene period.

2. Materials and Methods

2.1. Benthic Foraminifer Isotope Analysis

A total of 432 samples in Hole U1560A Cores 6–9H were analyzed for benthic foraminifer isotopes from 56 to 90 m Core Composite depth below Sea Floor (CCSF) (Figure 3), with a sampling resolution of averagely 6 cm. The CCSF depth is obtained by correlating the physical parameters and magnetic properties in Holes U1560A and U1560C (Coggon et al., 2024). The samples were sieved for the >63 μm fraction. 2–3 tests of *Cibicidoides*

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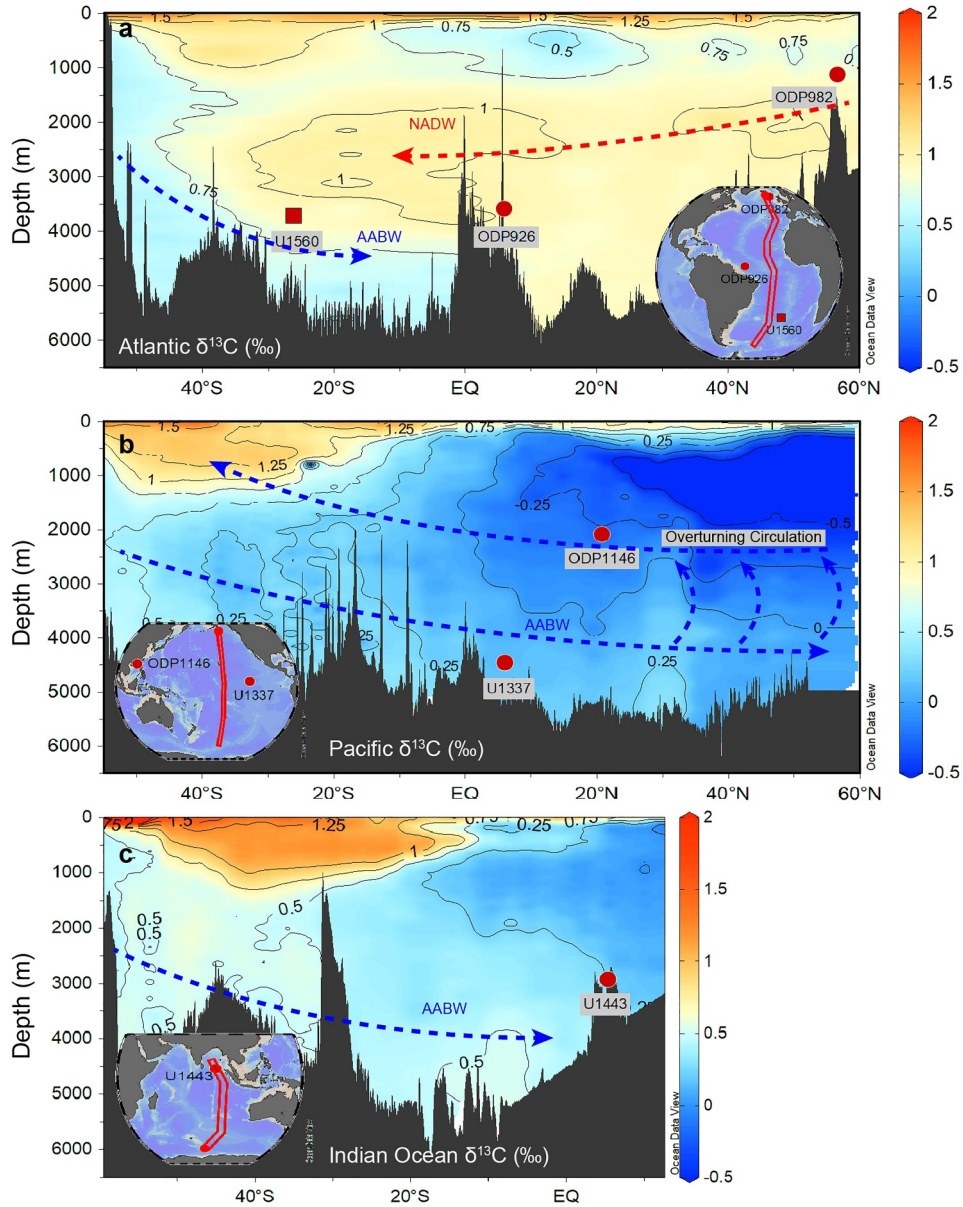


Figure 2. Maps showing the modern seawater dissolved inorganic carbon $\delta^{13}C$ (Global Ocean Data Analysis Project version 2.2023) (Lauvset et al., 2023) profiles in the Atlantic (a), Pacific (b), and Indian (c) Oceans, superimposed by ocean drilling sites (○) with the late Miocene $\delta^{13}C$ records (Bolton et al., 2022; Drury, Westerhold, et al., 2018; Hodell et al., 2001; A. E. Holbourn et al., 2018; Drury et al., 2017; Tian et al., 2018) (Table 1). This study presented new benthic foraminiferal isotopes in Hole U1560A (□) in the South Atlantic. The dashed lines and arrows sketch the flows of deep water mass, that is, North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW). Maps are produced by Ocean Data View (Schlitzer, 2024).

wuellerstorfi (>250 μm) (Figure S1 in Supporting Information S1) in each sample were treated by ultrasonic cleaning with alcohol, and dried in an oven at 60°C for 24 hr. Oxygen and carbon isotopic analysis was conducted using a Finigan MAT-253 mass spectrometer with a Kiel-IV carbonate device in Tongji University. Long-term measurements on Chinese national carbonate standard GBW04405 show an average analytical precision (1σ) of 0.07 and 0.04‰ for $\delta^{18}O$ and $\delta^{13}C$, respectively. The isotopic results are reported in the Vienna Pee Dee Belemnite scale (‰, VPDB) calibrated to the NBS19 standard. Replicated measurements were also carried out for more than 200 samples, showing no significant differences (Figure S2 in Supporting Information S1). The final benthic isotopes are reported as the average of replicates.

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Table 1
Representative ODP and IODP Sites for the Late Miocene Benthic Foraminiferal $\delta^{13}C$ Records With Well-Constrained Age Models in the Global Oceans

Site/Hole	Latitude (°N)	Longitude (°E)	Depth (m)	Location	References
982	57.517	−15.869	1,134	North Atlantic	Hodell et al. (2001) and Drury, Westerhold, et al. (2018)
926	3.737	−42.908	3,598	Tropical Atlantic	Drury et al. (2017)
U1560A	−30.403	−16.923	3,723	South Atlantic	This study
U1443	5.383	90.362	2,924	Indian Ocean	Bolton et al. (2022)
1146	19.457	116.273	2,092	Western Pacific	A. E. Holbourn et al. (2018)
U1337	3.833	−123.206	4,463	Eastern Pacific	Drury et al. (2017) and Tian et al. (2018)

Note. Notably, the records at Sites U1337 and 1146 are merged to obtain the CENOGRID, representing the deep ocean $\delta^{13}C$ in the open Pacific (De Vleeschouwer et al., 2020; Westerhold et al., 2020).

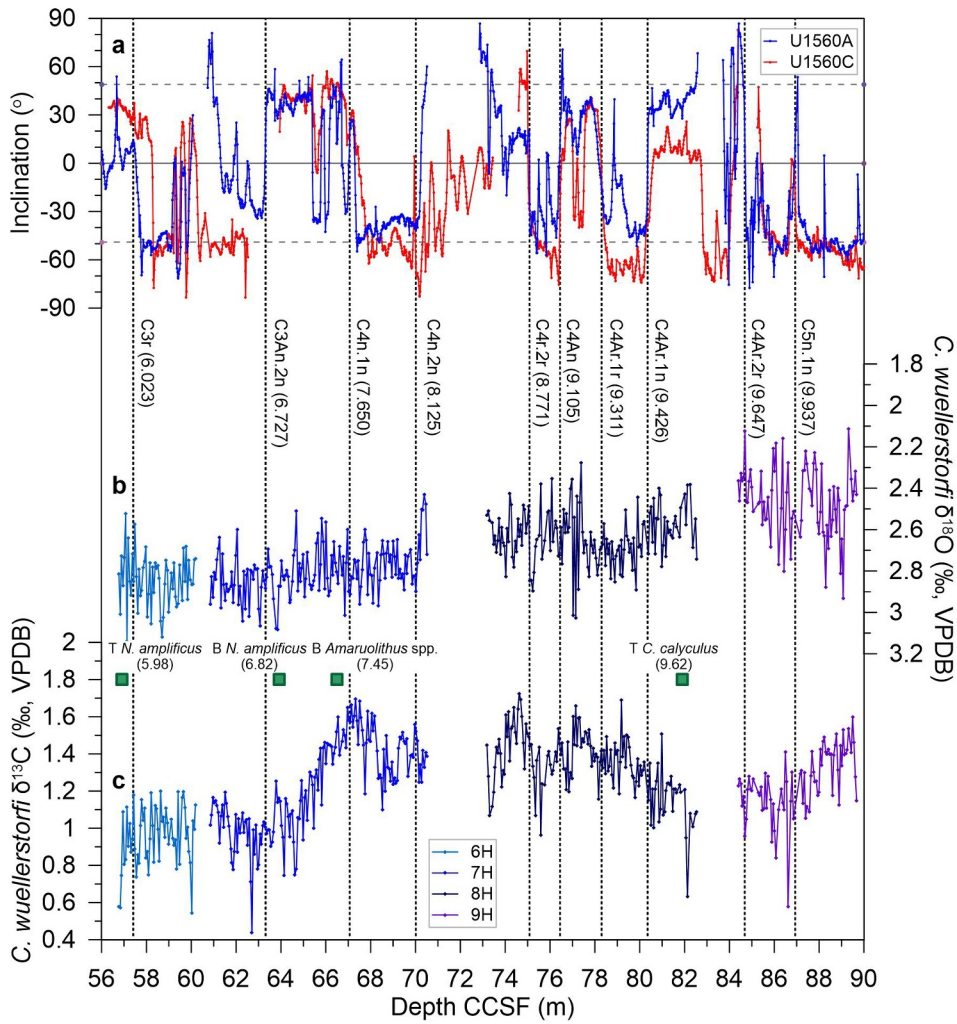


Figure 3. The shipboard inclination (a) (20 mT demagnetization) in Holes U1560A and U1560C from 56 to 90 m CCSF depth (Coggon et al., 2024), showing the geomagnetic reversals during the late Miocene, compared with the benthic foraminiferal $\delta^{18}O$ (c) and $\delta^{13}C$ (d) in Cores 6–9H in Hole U1560A. The squares (□) indicate the nannofossil bioevents investigated in Hole U1560A, showing no reversals with the paleomagnetism datum.

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Table 2
The Nannofossil and Shipboard Paleomagnetism Datum (Coggon et al., 2024) in Hole U1560A With Their Sample I.D., the Sample Above or Below, the Midpoint Depth CCSF, and the Assigned Geological Time Scale 2020 Ages (Gradstein & Ogg, 2020; Ogg, 2020; Raffi et al., 2020)

Datum	Core, section, interval (cm)	Sample above/below	Midpoint depth CCSF (m)	Age (Ma)
	EXP.395E U1560A	EXP.395E U1560A		
B. C3n.4n[a]	6H2, 30	6H2, 32	52.513	5.235
T. *Nicklithus amplificus*	6H5, 24–26	6H5, 17–19	56.918	5.98
B. C3r[a]	6H5, 70	6H5, 72	57.413	6.023
B. C3An.2n[a]	7H2, 108	7H2, 110	63.317	6.727
B. *Nicklithus amplificus*	7H3, 17–19	7H3, 23–25	63.936	6.82
B. *Amaruolithus* spp.	7H4, 131–133	7H4, 137–139	66.576	7.45
B. C4n.1n[a]	7H5, 34	7H5, 36	67.066	7.650
B. C4n.2n[a]	7H7,20	7H7, 30	70.016	8.125
δ^{13}C maximum[a]	8H2, 26–28	8H2, 20–22/8H2, 32–34	74.627	8.574
B. C4r.2r	8H2, 72	8H2, 74	75.087	8.771
B. C4An	8H3, 54	8H3, 56	76.407	9.105
δ^{13}C maximum[a]	8H3, 128–130	8H3, 122–124/8H3, 134–136	77.147	8.832
B. C4Ar.1r	8H4, 74	8H4, 76	78.117	9.311
δ^{13}C maximum[a]	8H5, 55–57	8H5, 50–52/8H5, 61–63	79.427	9.176
B. C4Ar.1n	8H5, 146	8H6, 0	80.352	9.426
T. *Catinaster calyculus*	8H7, 11–13	8H6, 144–146	81.917	9.62
B. C4Ar.2r[a]	9H1, 96	9H1, 98	84.684	9.647
B. C5n.1n[a]	9H3, 18	9H3, 20	86.914	9.937
B. C5n.2n[a]	10H1, 0	10H3, 148	96.302	11.056

Note. Additionally, the U1560A benthic foraminiferal δ^{13}C in Core 8H are considered as tie-points, by correlating the δ^{13}C maxima to the CENOGRID (Westerhold et al., 2020). T. and B. indicate top and base of bioevent/chronology, respectively. [a]Tie-points used to construct the age-depth model in Hole U1560A.

2.2. Age Model in Hole U1560A

The shipboard nannofossil biostratigraphy and magnetostratigraphy show that the analyzed samples belong to the late Miocene period (Coggon et al., 2024). Here smear slides for each sample were prepared to refine the nannofossil datum. A total of four nannofossil bioevents were identified as the top of (T.) *Nicklithus amplificus*, the base of (B.) *N. amplificus*, B. *Amaruolithus* spp., and T. *Catinaster calyculus* (Table 2; Figure S3 in Supporting Information S1). These events are assigned to 5.89, 6.82, 7.45, and 9.62 Ma, respectively, according to Geological Time Scale 2020 (GTS2020) (Gradstein & Ogg, 2020; Raffi et al., 2020). The upper three bioevents, that is, T. and B. *N. amplificus*, B. *Amaruolithus* spp., agree with the paleomagnetism datum indicated by the bases of C3r, C3An.2n, and C3Br.2r, respectively (Figure 3; Table 2), suggesting that these paleomagnetism chronologies can provide strong constraints on the age-depth model.

The paleomagnetism chronologies in Holes U1560A and U1560C are identified and correlated to the Geomagnetic Polarity Time Scale (Ogg, 2020) (GPTS20), and considered as tie-points (Figure 4; Table 2). Then, depths are linearly interpolated between the tie-points to construct the age model. The present age model shows that the U1560A benthic δ^{13}C can capture the general pattern of the δ^{13}C of Cenozoic Global Reference benthic foraminifer carbon and oxygen Isotope Data set (CENOGRID) (Westerhold et al., 2020), however, with significant divergences with respect to the CENOGRID during 8.6–9.6 Ma, covering the CCSF depth from 73 to 83 m in Core 8H (Figure S5 in Supporting Information S1). The late Miocene CENOGRID is obtained by merging the benthic isotopes of *Cibicidoides* spp. at IODP Site U1337 (Drury et al., 2017; Tian et al., 2018) in the equatorial Eastern Pacific and ODP Site 1146 (A. E. Holbourn et al., 2018) in the South China Sea in the tropical Western Pacific (De Vleeschouwer et al., 2020; Westerhold et al., 2020) (Figure S4 in Supporting Information S1). The CENOGRID serves as a high-definition stratigraphic reference over the Cenozoic. Furthermore, the age model shows remarkably dense data points between 9.1 and 9.6 Ma, but sparse points between 8.6 and 9.1 Ma,

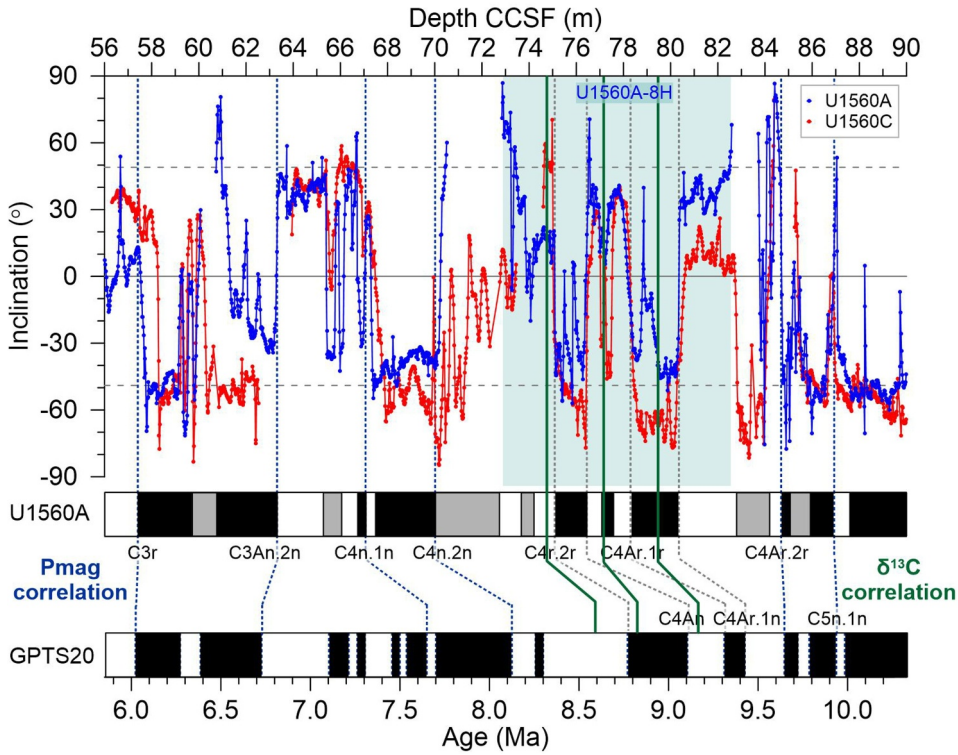


Figure 4. The paleomagnetism (Pmag) chronologies identified based on the shipboard inclination (20 mT demagnetization) in Holes U1560A and U1560C (Coggon et al., 2024), correlated to GPTS20 (Ogg, 2020). In Hole U1560A Core 8H (the shading), the measured benthic $\delta^{13}C$ are correlated to the CENOGRID (Westerhold et al., 2020) to construct the age model which shows divergences of 100–300 kyr with respect to the Pmag-based age model in Core 8H.

indicating that linear sedimentation rate (LSR) in the lower part is 4- to 5-fold greater than the upper part of Core 8H in Hole U1560A (Figure S5 in Supporting Information S1). The divergences of benthic $\delta^{13}C$ and unusual LSR may result from that the paleomagnetism chronologies in Core 8H (i.e., the bases of C4r.2r, C4An, C4Ar.1r, C4Ar.1n) do not perfectly match with GPTS20. Because, there are undetermined geomagnetic intervals in the bottom and top of Core 8H, so that the identified chronologies in Hole U1560A were not one-to-one correlated to GPTS20 (Figure 4). Additionally, a "normal" geomagnetic interval can be identified between C4r.An and C4Ar.1n at 77–77.5 m CCSF depth in Holes U1560A and U1560C, however, it does not present in GPTS20 (Figure 4).

In the present study, we aim to figure out the benthic $\delta^{13}C$ differences in the global oceans, therefore, the tie-points of measured benthic foraminiferal $\delta^{13}C$ maxima at 74.627, 77.147, and 79.427 m CCSF depth were correlated to the CENOGRID $\delta^{13}C$ at 8.574, 8.832, and 9.176 Ma, respectively (Table 2; Figures 4 and 5), rather than the paleomagnetism datum in Core 8H. Accordingly, the U1560A benthic $\delta^{13}C$ show significant 405-kyr cycles in the long-eccentricity band (Figure S6 and S7 in Supporting Information S1). The investigated cores span an age from ~5.9 to 10.3 Ma, with a flattened LSR of averagely ~7.89 mm kyr^{-1} (Figure 5). Although the benthic isotopes in Hole U1560A were not tuned to an astronomical target due to the relatively low resolution and discontinuous isotope records, the $\delta^{13}C$ in Hole U1560A here is suitable for analyzing a long-term change in deep ocean $\delta^{13}C$ over million years.

The $\delta^{13}C$-based age model would cause significant divergences of 100–300 kyr in the bases of C4r.2r, C4An, C4Ar.1r, and C4Ar.1n in Hole U1560A-8H with respect to GPTS20, which can primarily result from that the geomagnetism correlation here may not be ideal. Another possibility for explaining the age discrepancy could be that the magnetostratigraphy and benthic foraminiferal isotope stratigraphy originate in two types of independent materials. The late Miocene GPTS20 chronologies are from the magnetostratigraphy of the deep marine sediments outcropped at Monte dei Corvi, northern Italy, showing significant sapropel cycles (Hüsing et al., 2007,

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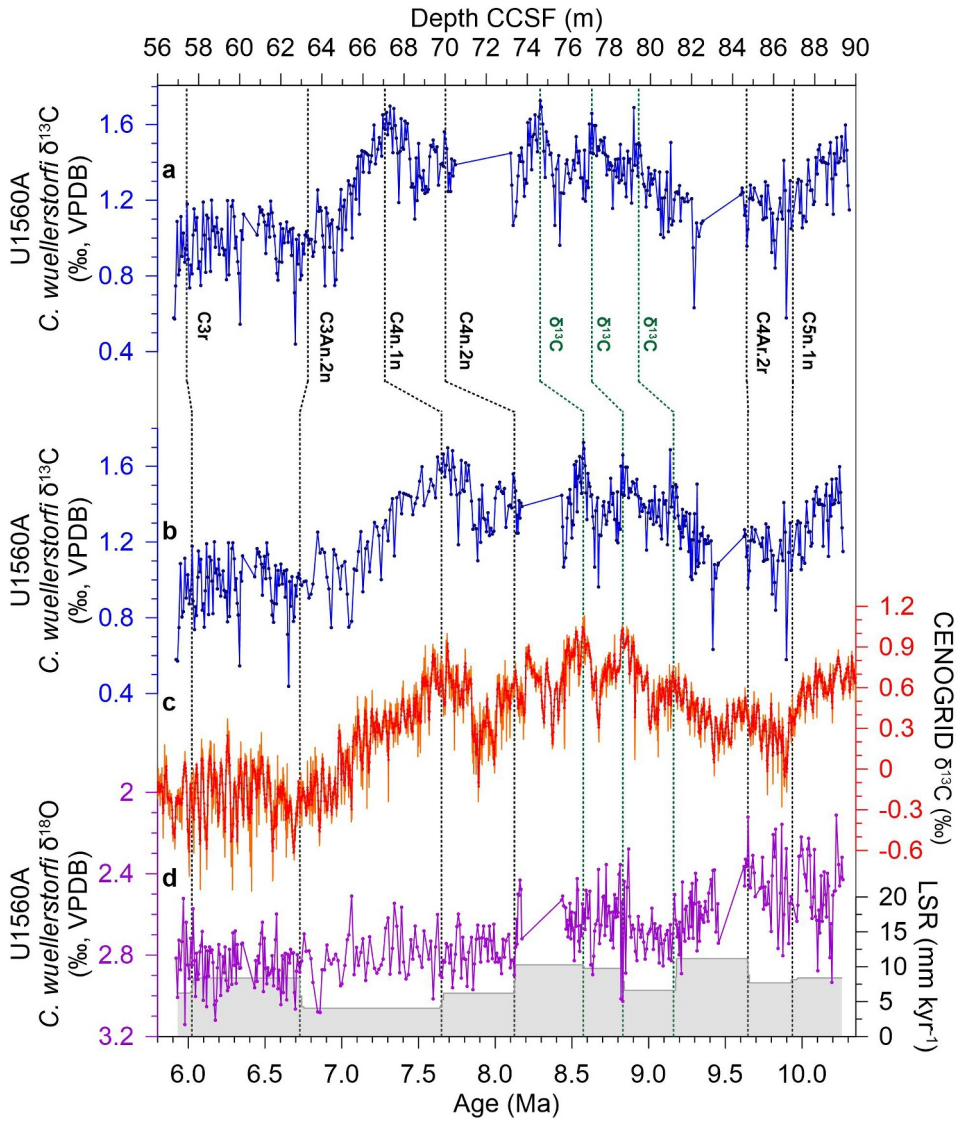


Figure 5. The Hole U1560AC. *wuellerstorfi* $\delta^{13}C$ plotted with CCSF depth (a), and the *C. wuellerstorfi* $\delta^{13}C$ (b) and $\delta^{18}O$ (d) plotted with age, compared with the late Miocene CENOGRID $\delta^{13}C$ (c) (Westerhold et al., 2020). The dashed lines indicate the tie-points of paleomagnetism chronology and $\delta^{13}C$ (Table 2) which are used to construct the age-depth model in Hole U1560A. The "$\delta^{13}C$" tie-points are the measured benthic $\delta^{13}C$ in Hole U1560A correlated to the CENOGRID (b, c); (d) The bar-plot indicates linear sedimentation rate (LSR) based on the age model.

2009). The sedimentary cycles are tuned to obliquity and precession (Laskar et al., 2004), with constraints from independent radioisotope dating after 7.6 Ma (Hüsing et al., 2007, 2009), however, with no benthic foraminiferal isotopes for cross-comparison. The CENOGRID before 8.2 Ma of the late Miocene is from the marine sediments at ODP Site 1146 in the South China Sea (tropical Western Pacific), unfortunately, with no magnetostratigraphy during this interval (A. E. Holbourn et al., 2018). Interestingly, a recent study reanalyzed the astronomical-tuning age at Site U1337, and suggests a potential age divergence of ~300 kyr with respect to the CENOGRID during the early Miocene period (Hilgen, 2025). This means that the present Astronomical Time Scale framework still needs to be refined in future studies (Westerhold et al., 2024).

2.3. Benthic Foraminiferal $\delta^{13}C$ Differences

Differences between the benthic foraminiferal $\delta^{13}C$ records (Table 1) are calculated to show the evolution of global deep ocean $\delta^{13}C$ pattern over the late Miocene. The benthic $\delta^{13}C$ records are all from *Cibicidoides* spp.,

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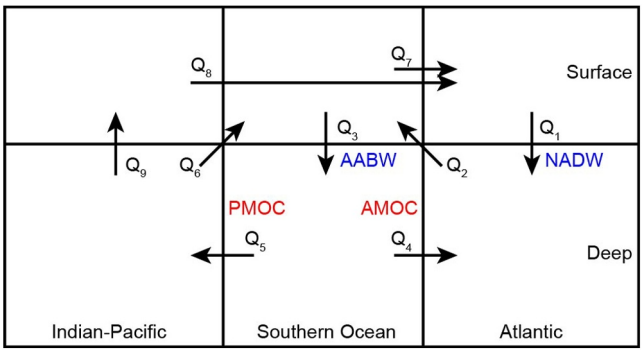


Figure 6. Physical configuration of the carbon cycle box-model designed here. The model has six boxes, including Surface and Deep Atlantic, Southern Ocean, and Indian-Pacific Boxes. The arrows indicate water transport between the boxes referring to Talley (2013): $Q_1 = 19$ Sv; $Q_2 = 26$ Sv; $Q_3 = 29$ Sv; $Q_4 = 7$ Sv, $Q_5 = 22$ Sv; $Q_6 = 21$ Sv; $Q_7 = 18$ Sv; $Q_8 = 1$ Sv; $Q_9 = 1$ Sv (1 Sv = 10^6 m^3 s^{-1}). Here Q_1 and Q_3 denote NADW and AABW formation, respectively. $Q_{2,3,4}$ and $Q_{3,5,6}$ circulation loops represent the lower meridional overturning circulation cells in the Atlantic and Indian-Pacific sectors, that is, AMOC and PMOC, respectively.

and thus their isotopic values have not been corrected (Westerhold et al., 2020). First, the δ^{13}C time-series are equally interpolated with a resolution of 10 kyr. To reduce the influences from age inconsistency between the benthic δ^{13}C records, the δ^{13}C differences are calculated between the locally estimated scatterplot smoothing (loess) using weighted linear least squares and a 2-degree polynomial model with a window of 40 kyr (Figure S8 in Supporting Information S1). The smoothing was conducted using the Matlab code "smooth." The smoothed curve can be regarded as a regression of the time-series, then, σ can be obtained based on the errors which are normally distributed, $\boldsymbol{y}_{\text{data}} - \boldsymbol{y}_{\text{smooth}} = N(0, \sigma^2)$ (Figure S8 in Supporting Information S1). Finally, σ of the δ^{13}C difference between two time-series can be calculated, $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$, and $\pm 1\sigma$ indicates the 68% confidence interval.

2.4. A Diagnostic Carbon Cycle Box-Model

Here a simple carbon cycle box-model is developed to diagnose the drivers of global deep ocean δ^{13}C pattern during the late Miocene. The model is designed with six boxes which indicate Surface and Deep Atlantic, Surface and Deep Southern Ocean, Surface and Deep Indian-Pacific (Figure 6). The physical configuration and water transport between the boxes can be characterized by the overturning cells in the Atlantic and Indian-Pacific sectors, following the global overturning circulation closure (Talley, 2013). The NADW and AABW formation is denoted by the water transport from Surface to Deep Boxes in Atlantic and the Southern Ocean, respectively. Circulation loops $Q_{2,3,4}$ and $Q_{3,5,6}$ (Figure 6) represent the deep meridional overturning circulation (MOC) cells in the Atlantic and Indian-Pacific sectors, respectively, that is, AMOC and PMOC.

The biological processes include organic carbon fixation via primary productivity and its degradation in the water column. We assume 1% of the total carbon fixation buried in marine sediments which must be balanced by riverine carbon input. The changes in dissolved inorganic carbon concentration ([DIC], mol m^{-3}) with time is

$$\frac{\mathrm{d[DIC]}}{\mathrm{d}t} = \frac{1}{V}\left(\sum Q_i[\mathrm{DIC}]_i + F_{\text{fix}} + F_{\text{river}}\right), \tag{1}$$

where V (m^3) is the volume of a box; $\sum Q_i[\mathrm{DIC}]_i$ indicates the DIC changes due to flows or water transport (Q, unit in Sv, 1 Sv = 10^6 m^3 s^{-1}); F_{fix} (mol carbon s^{-1}) indicates net carbon fixation or degradation; F_{river} (mol carbon s^{-1}) indicates the riverine carbon input. The change in DIC isotopic composition (δ_c) with time is

$$\frac{\mathrm{d}\delta_c}{\mathrm{d}t} = \frac{1}{V * [\mathrm{DIC}]}\left(\sum Q_i[\mathrm{DIC}]_i(\delta_c)_i + F_{\text{fix}}(\delta_c - \epsilon_p) + F_{\text{river}}\delta_{\text{river}}\right), \tag{2}$$

where δ_{river} indicates the riverine carbon δ^{13}C; ϵ_p is the fractionation factor through photosynthesis.

The box-model neglects the calcium carbonate production, dissolution, burial, and an atmosphere box with the associated sea-air CO_2 exchange. The model was run with the modern boundary conditions to attain steady state. The steady state primary productivity (PP), [DIC], and seawater δ^{13}C can be found in Table S1 in Supporting Information S1. The δ^{13}C in Deep Atlantic, Deep Southern Ocean, and Deep Indian-Pacific Boxes are 0.51‰, −0.05‰, and −0.29‰, respectively, with a δ^{13}C difference of 0.8‰ between Deep Atlantic and Indian-Pacific Boxes ($\Delta\delta^{13}C_{\text{Atl.}-\text{Pac.}}$). The patterns of deep δ^{13}C and [DIC] (Table S1 in Supporting Information S1) are close to the modern oceans (Figure 2), indicating that the box-model is able to diagnose the controlling factors on global deep δ^{13}C pattern. Based on the steady state conditions, a series of sensitivity tests were carried out by changing the ocean PP, the flows between the boxes, the riverine carbon δ^{13}C to decipher the potential drivers of global deep ocean δ^{13}C pattern.

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3. Results and Discussion

3.1. Benthic Foraminiferal Isotopes in Hole U1560A

The U1560A benthic $\delta^{18}O$ show a long-term and step-wise positive shift, with an oscillation range of ~0.5‰ (Figure 4; Figure 7). Two major periods of the $\delta^{18}O$ positive shift are observed during ~7.7–8.2 Ma and ~9.2–9.6 Ma.

The U1560A benthic $\delta^{13}C$ show a strong negative excursion/shift by ~0.6‰ (from 1.6 to 1.0‰) during 7.0–7.6 Ma (Figures 4 and 7). After this period, the average benthic $\delta^{13}C$ is ~1‰, close to the modern deep seawater $\delta^{13}C$ in the South Atlantic (Figure 2). The negative excursion can be assigned to the LMCIS. The onset of LMCIS observed in Hole U1560A at 7.6 Ma agrees with the late Miocene CENOGRID $\delta^{13}C$ derived from Site U1337 in the open Pacific Ocean (Drury et al., 2017; Westerhold et al., 2020). Comparatively, the duration of LMCIS observed in the CENOGRID seems to be longer and lasted for 1 Myr, with a greater magnitude of negative shift by ~0.8‰ (from 0.6‰ to −0.2‰). The apparently shorter duration of LMCIS observed in Hole U1560A cannot be attributed to the age model, because the paleomagnetism chronologies agree with the nannofossil bioevents at ~6.8 Ma and ~7.5 Ma (Table 2), suggesting confident age controls. The U1560A benthic $\delta^{13}C$ show significant 405-kyr cycles (Figure S6 in Supporting Information S1). In the 405-kyr band, $\delta^{13}C$ maxima are corresponding to eccentricity minima, suggesting an orbital forcing on the deep ocean $\delta^{13}C$ (Figure S7 in Supporting Information S1).

3.2. General Patterns of the Late Miocene Ocean Carbon Cycle

The inter-basin comparison of benthic $\delta^{13}C$ in global oceans is shown in Figure 7, including Site 982 in the North Atlantic (Drury, Westerhold, et al., 2018; Hodell et al., 2001), Site 926 in the equatorial Atlantic (Drury et al., 2017), Hole U1560A in the South Atlantic (this study), Site U1443 in the Indian Ocean (Bolton et al., 2022), Site 1146 in the South China Sea (tropical Western Pacific) (A. E. Holbourn et al., 2018), and Site U1337 in the equatorial Eastern Pacific (Drury et al., 2017; Tian et al., 2018) (Table 1). The benthic $\delta^{13}C$ at Site 1146 are adjusted to Site U1337 (Figure S4 in Supporting Information S1) to obtain the late Miocene $\delta^{13}C$ composited curve (i.e., CENOGRID), here noted as W2020, representing the deep ocean $\delta^{13}C$ in the open Pacific (De Vleeschouwer et al., 2020; Westerhold et al., 2020). We note that some published $\delta^{13}C$ records have not been shown, because their isotopic values and locations are very close to those already cited here. For example, the pattern of benthic $\delta^{13}C$ at Site 999 in the Caribbean Sea is identical to Site 926 in the equatorial Atlantic, although the benthic $\delta^{13}C$ at Site 999 is overall ~0.5‰ more negative than Site 926 (Bickert et al., 2004).

A general pattern to character the late Miocene deep ocean $\delta^{13}C$ is the secular 405-kyr long-eccentricity carbon cycles superimposed by the LMCIS (Figure 7). The 405-kyr cycles in benthic $\delta^{13}C$ can be identified in all the representative records over the late Miocene. The eccentricity-paced ocean $\delta^{13}C$ has long been recognized through the Cenozoic period (since 65 Ma) (De Vleeschouwer et al., 2020; Palike et al., 2006; Westerhold et al., 2020). For example, the periodic $\delta^{13}C$ maxima in the long-eccentricity cycles during the middle Miocene are know as the "Monterey Excursion" (A. Holbourn et al., 2007). It is suggested that the periodicities in $\delta^{13}C$ can result from the expansion and contraction of land carbon reservoir and/or the changes in ocean carbon biogeochemical processes (De Vleeschouwer et al., 2020; F. Liu et al., 2024; W. Ma et al., 2011; X. Ma et al., 2022; Palike et al., 2006).

Eccentricity can modulate the amplitude of precession and in turn control the seasonal insolation and the low-latitude hydrological cycles. At high eccentricity, a stronger hydrological cycle will cause increased terrestrial carbon which holds more ^{12}C input to ocean, leading to negative seawater $\delta^{13}C$ (De Vleeschouwer et al., 2020). A "coral reef" hypothesis can be another candidate mechanism for the eccentricity forcing. The core statement is that the burial of shelf carbonates controlled by sealevel plays a key role for ocean-atmosphere carbon cycling (Berger, 1982). A revised coral reef mechanism for the Miocene period can be that at high eccentricity a stronger hydrological cycle will induce greater export productivity of shelf calcium carbonate in low-latitudes, which leads to an overall increase in total inorganic to organic carbon burial ratio, therefore, lowering the oceanic $\delta^{13}C$ (W. Ma et al., 2011; X. Ma et al., 2022).

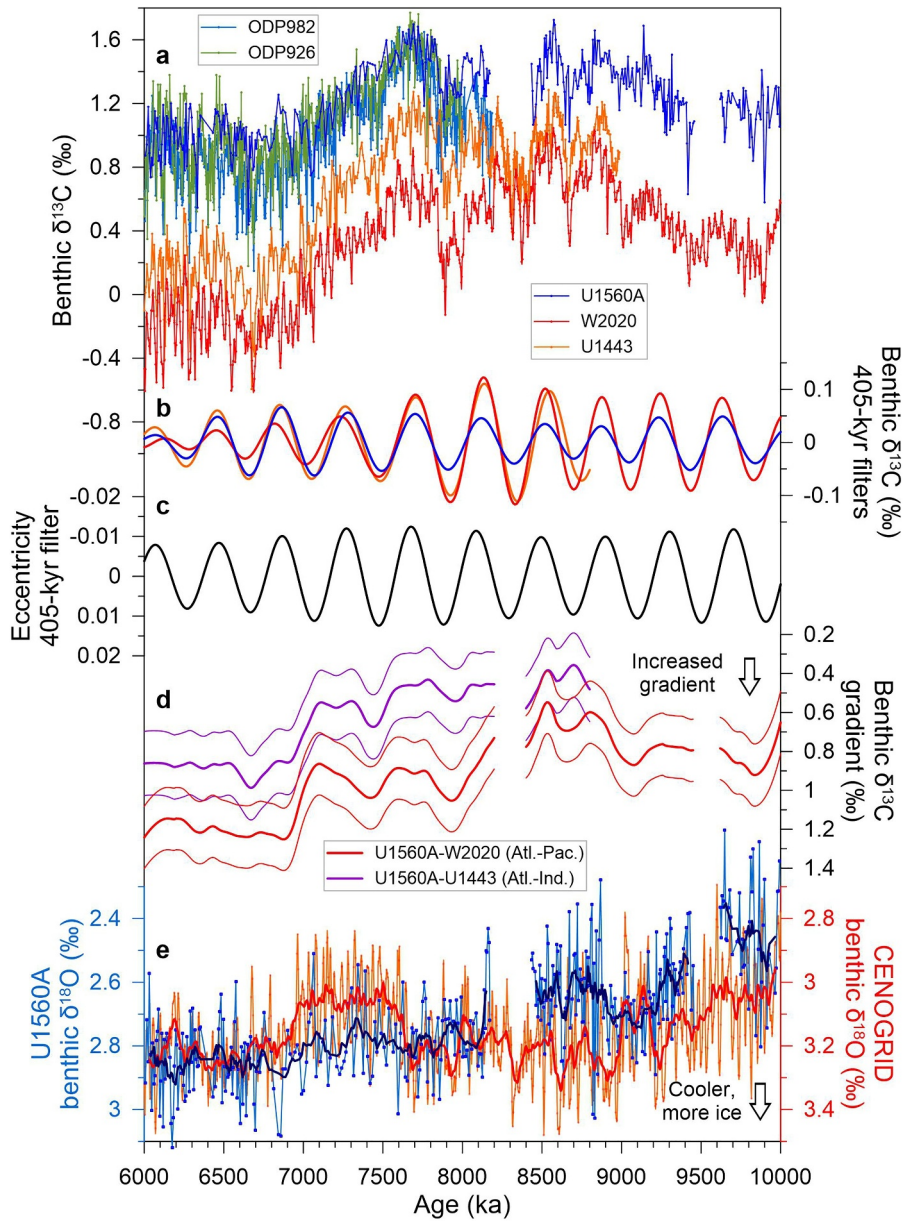


Figure 7. (a) Comparison of the benthic foraminiferal $\delta^{13}C$ in Hole U1560A in the South Atlantic (this study), Site 982 in the North Atlantic (Drury, Westerhold, et al., 2018; Hodell & Venz-Curtis, 2006), Site 926 (Drury et al., 2017) in the equatorial Atlantic, Site U1443 in the Indian ocean (Bolton et al., 2022), and the Pacific $\delta^{13}C$ (Westerhold et al., 2020), here noted as W2020, obtained from Sites 1146 (A. E. Holbourn et al., 2018) in the South China Sea (tropical Western Pacific) and U1337 (Drury et al., 2017; Tian et al., 2018) in the equatorial Eastern Pacific over the late Miocene; (b–c) The 0.00247 ± 0.00047 cycles kyr^{-1} (405-kyr) gauss band-pass filtering of benthic $\delta^{13}C$ of U1560A (South Atlantic), U1443 (Indian Ocean), W2020 (Pacific), and eccentricity (Laskar et al., 2004). The filtering analyses were conducted using the Acycle Matlab (Li et al., 2019); (d) Differences of the benthic $\delta^{13}C$ (bold lines) between the Atlantic and Indian, Pacific Oceans, showing increased basinal $\delta^{13}C$ gradient over the late Miocene. Fine lines indicate the $\pm 1\sigma$ interval; (e) The benthic $\delta^{18}O$ in Hole U1560A (South Atlantic) compared with the CENOGRID $\delta^{18}O$ (Westerhold et al., 2020). Bold lines indicate the running average smoothing. Increased benthic $\delta^{18}O$ can indicate cooling of deep water or greater land ice-sheet volume.

The LMCIS can associate with the expansion of C_4-grassland over the late Miocene (e.g., Cerling et al., 1993), suggesting a strong perturbation of land ecosystem. This point of view originates in the fact that there is a positive excursion in *n*-alkane $\delta^{13}C$ signatures discovered in Pakistan paleosol since ~8 Ma, indicating the C_4-grassland expansion in South Asia (Freeman & Colarusso, 2001; Karp et al., 2018; Polissar et al., 2021). A possible driving

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mechanism is suggested as the land-ocean mass balance of carbon, that is, C_4-plants can sequester more heavy carbon (^{13}C) in land, leading to more negative oceanic $\delta^{13}C$ (Katz et al., 2005; Kump & Arthur, 1999). Based on this framework, Du et al. (2022) have simulated the LMCIS using a carbon cycle box-model, and a negative excursion in river-to-ocean carbon isotopic signature is prescribed as a forcing. Another possible mechanism could be a net transfer of terrestrial organic carbon with lighter isotopic signatures to ocean, since the land biomass with higher grassland covering is supposed to have lower carbon density (Diester-Haass et al., 2006). Independent evidence showing a rapid increase in ocean phosphate concentration after ~7 Ma (across the LMCIS) reconstructed from hydrothermal sediments may support this idea, because the terrestrial organic carbon erosion and its input to ocean are accompanied by the nutrient element releasing (Wang et al., 2022).

It is also argued that the land vegetation change was rather gradual and globally asynchronous, and thus such a cause of LMCIS may be debated (Drury et al., 2017). For example, the onset of positive shift in *n*-alkane $\delta^{13}C$ which indicates the C_4-grassland expansion is observed at 10 Ma in East Africa (Uno et al., 2016), much earlier than South Asia. A positive shift in isotopic signatures of black carbon are also recorded in the Japan Sea, revealing the onset of C_4-grassland expansion at ~5 Ma in the northern Central Asia (Shen et al., 2018). It has long been suggested that a decline in atmospheric CO_2 may be a driver for the late Miocene C_4-plants evolution, however, this hypothesis may be problematic due to the globally asynchronous land C_3/C_4 turnover and the relative stable atmospheric CO_2 concentration (CenCO2PIP-Consortium, 2023) (Figure 1d). It is more plausible that the expansion of C_4-plants can be favored by a long-term aridity over the middle to late Miocene associated with the tectonic activities, for example, the Tibetan Plateau uplift (Miao et al., 2012).

3.3. Intensified Basinal $\delta^{13}C$ Gradient Over the Late Miocene

Here the comparison of global benthic $\delta^{13}C$ shows that there is a step-wise increased deep ocean $\delta^{13}C$ difference between Hole U1560A in the South Atlantic and Site U1443 in the Indian Ocean, the Pacific W2020 $\delta^{13}C$ (Figure 7d). Two periods are responsible for the intensified $\delta^{13}C$ difference. The first period occurred during 7.8–8.4 Ma, with increased $\delta^{13}C$ difference by 0.4–0.5‰ between the South Atlantic and Pacific Ocean. The second period was at around 7 Ma across the LMCIS, and the $\delta^{13}C$ difference has increased by ~0.4 ‰ between the South Atlantic and Indian, Pacific Oceans (Figure 7d). Furthermore, the increased $\delta^{13}C$ differences between Sites 926, 982 and the Pacific W2020 are also observed across the LMCIS (Figure S9 in Supporting Information S1), suggesting an intensified basinal deep ocean $\delta^{13}C$ gradient between the Atlantic and Pacific Oceans. Additionally, filtering analyses show that the $\delta^{13}C$ maximum in the 405-kyr cycles centered at ~6.8 Ma is less prominent in the Pacific W2020 curve (Figures 7a–7c) than Hole U1560A, which may interpret the apparently shorter LMCIS duration observed in Hole U1560A. Indeed, the increased deep ocean $\delta^{13}C$ difference across the LMCIS between the Atlantic and Pacific Oceans has been observed previously (Bickert et al., 2004; Hodell & Venz-Curtis, 2006). Here our new benthic foraminiferal isotope record in the South Atlantic reveals that such a change in deep ocean $\delta^{13}C$ pattern may occur over the late Miocene. Possible causes for this ocean carbon cycle change may include various physical and biogeochemical processes. Here we try to figure out this problem using the simple diagnostic carbon cycle box-model developed here (Figure 6).

3.3.1. Ocean Primary Productivity

Increased ocean PP was analyzed using the diagnostic box-model, showing that:

1. An increase of PP in the Southern Ocean by 20%–50% would cause intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ by ~0.08–0.2‰ (Figure 8a). It takes ~3,000 years for the system to arrive at steady state. The intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ primarily results from the increased $\delta^{13}C$ in Deep Atlantic Box, as enhanced carbon fixation discriminates against more ^{13}C in the surface Southern Ocean that is transported to the Atlantic boxes through Q_7 (Figure 6).
2. An increase of PP in Surface Indian-Pacific Box by 20%–50% would cause intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ by ~0.05–0.1‰ (Figure 8c). It takes a longer time, ~7,000 years, to arrive at steady state. The intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ is primarily caused by a greater amount of organic carbon degradation that releases more ^{12}C in Deep Indian-Pacific Box. In this case, the $\delta^{13}C$ gradient between the deep Southern Ocean and Indian-Pacific boxes would not change significantly.
3. An increase of PP by 20%–50% in Surface Atlantic Box would not greatly affect the deep $\delta^{13}C$ and its gradient in the global oceans (Figure 8b), because the $\delta^{13}C$ perturbation by PP is weak due to the relatively small

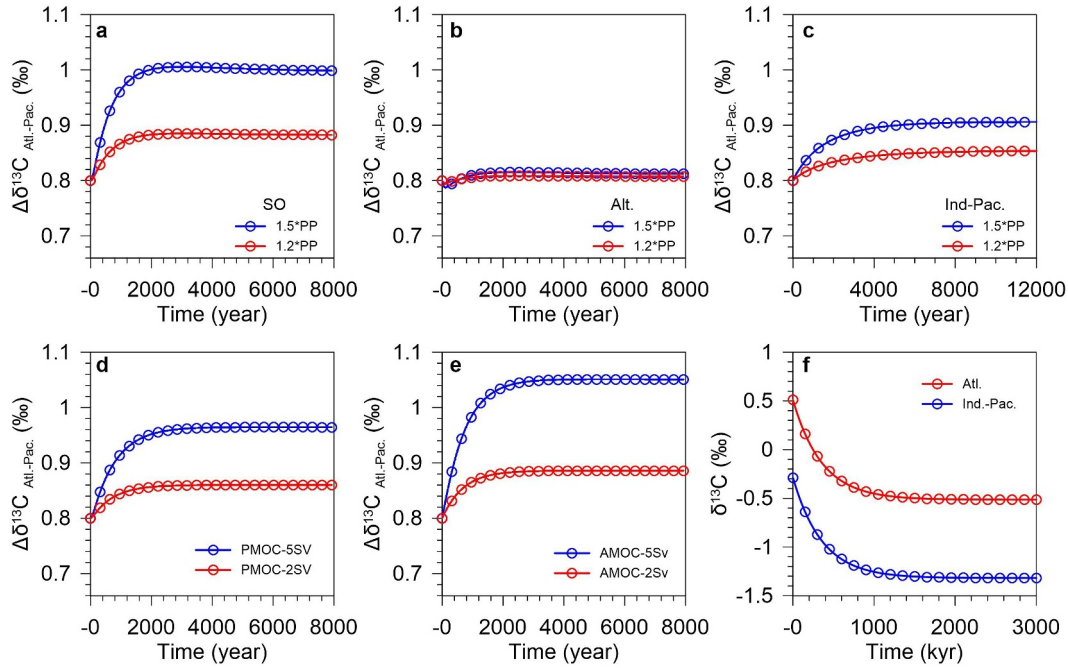


Figure 8. Sensitivity tests showing the changes in $\delta^{13}C$ difference between Deep Atlantic and Indian-Pacific Boxes ($\Delta\delta^{13}C_{Atl.-Pac.}$) (a–e) and the $\delta^{13}C$ evolution in Deep Atlantic and Indian-Pacific Boxes with decreased riverine carbon $\delta^{13}C$ by 1‰ (f). (a–c) show the results from 1.2- and 1.5-fold primary productivity in Surface Southern Ocean (SO), Atlantic (Atl.), and Indian-Pacific (Ind-Pac.) Boxes, respectively. (d, e) Show increased $\Delta\delta^{13}C_{Atl.-Pac.}$ due to sluggish or weakening of ocean meridional overturning circulations. PMOC- and AMOC-*x* Sv here indicate the reduced water flows in the $Q_{3,5,6}$ and $Q_{2,3,4}$ circulation loops, respectively.

volume of the Atlantic boxes and vigorous water transport between them. That is, the perturbation is muted, since the water columns are well-ventilated or mixed in the North Atlantic due to the formation of deep waters, comparatively, no such deep waters are formed in the North Pacific and Indian Ocean (Talley, 2013).

Increased ocean PP during the late Miocene has long been suggested and referred to as the "biogenic bloom" during 6–8 Ma (Diester-Haass et al., 2005, 2006). However, the biogenic bloom is globally diachronous; for example, the peak of biogenic bloom centered at 6.8 and 7.4 Ma in the South Pacific and the eastern equatorial Pacific, respectively (Diester-Haass et al., 2006). Assuming an intensified basinal $\delta^{13}C$ gradient was caused by increased ocean PP, the $\Delta\delta^{13}C_{Atl.-Pac.}$ would likely return to a smaller value shortly after the end of biogenic bloom. In fact, the basinal $\delta^{13}C$ gradient between the Atlantic and Pacific Oceans has generally maintained its value after the LMCIS during the late Miocene (Figure 7). Furthermore, increased ocean PP would cause enhanced surface-to-deep $\delta^{13}C$ gradient. This contradicts the observations that there were no significant changes in the surface-to-deep $\delta^{13}C$ gradient, as revealed by approximately unchanged planktonic and benthic foraminiferal $\delta^{13}C$ difference at Site 1146 (South China Sea) (A. E. Holbourn et al., 2018) and Site U1338 (equatorial Eastern Pacific) (Drury, Lee, et al., 2018) over the late Miocene. Therefore, these observations may not support increased ocean PP as a primary driver for the intensified basinal $\delta^{13}C$ gradient.

3.3.2. Ocean Overturning Circulation

Here sluggish or weakening of Meridional Overturning Circulation (MOC) in Atlantic and Indian-Pacific Boxes (i.e., AMOC and PMOC) can be implemented by reducing the flows in circulation loops $Q_{3,5,6}$ and $Q_{2,3,4}$, respectively (Figure 6). After changing the water transport, [DIC] and $\delta^{13}C$ can arrive at steady state in ~3000 years (Figures 8d and 8e). The results show that reducing the flows of PMOC by 2–5 Sv will cause intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ by 0.06–0.16‰ (Figure 8d). Similar results are shown by reducing the flows of AMOC by 2–5 Sv, which will cause intensified $\Delta\delta^{13}C_{Atl.-Pac.}$ by 0.08–0.25‰ (Figure 8e). The intensified gradient is caused by increased deep Atlantic $\delta^{13}C$ and decreased deep Indian-Pacific $\delta^{13}C$, together with decreased deep

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Atlantic [DIC] and increased deep Indian-Pacific [DIC]. The box-model reveals that more DIC with negative $\delta^{13}C$ signature could be trapped in the deep Indian and Pacific Oceans, when the meridional overturning circulation was weakened.

The reorganization of ocean circulation can be linked to the degree of glaciation or the sea-ice content in high-latitude oceans, where the deep waters are formed. We may expect enhanced Antarctic glaciation or sea-ice covering in the Southern Ocean, corresponding to the late Miocene cooling of high-latitude sea surface temperature (Herbert et al., 2016). A drilling site with records of various geological and geophysical parameters in the eastern Antarctic continental shelf reveals expanded ice-sheet covering since the late Miocene (Gulick et al., 2017). An ocean drilling record of ice-rafted debris contents in the Indian Ocean sector of Southern Ocean also suggests the occurrence of large icebergs since the late Miocene (Williams et al., 2010). These geological records may support enhanced Antarctic glaciation during the late Miocene.

The expansion and contraction of Antarctic sea-ice can be a modulator for deep ocean overturning circulation. For an instance of the Last Glacial Maximum (LGM, ~21 ka) condition, expanded sea-ice can serve as a barrier for the Southern Ocean upwelling and its energy exchange with atmosphere (Ferrari et al., 2014). The isopycnal that divides the upper and lower overturning cells would become shallow, leading the deep overturning cell to be more isolated (Ferrari et al., 2014). Meanwhile, this will cause a northward expansion of the southern-sourced Antarctic bottom water (i.e., AABW), and increased stratification between the deep and intermediate waters in the ocean basins (Ferrari et al., 2014). A similar condition may occur as a result of enhanced Antarctic glaciation during the late Miocene. For example, a greater stratification between the intermediate and deep ocean layers over the late Miocene could be indicated by increased seawater neodymium isotope difference between Site U1483 (4,600 m depth) and Site 807 (2,800 m depth) in the Pacific Ocean (Kender et al., 2018; Le Houedec et al., 2016) (Figure 9a), suggesting enhanced isolation of the deep overturning cell in the Pacific. Considering an expansion of the southern-sourced deep water, the residence time of deep water mass would become greater, which may ultimately lead to sluggish or weakening of meridional overturning circulation. This kind of mechanism can be supported by an example of the radiocarbon data in the South Pacific which shows a greater reservoir age in the deep water column during the LGM with respect to Holocene (Skinner et al., 2015).

Our new benthic foraminiferal $\delta^{18}O$ in Hole U1560A in the South Atlantic show a long-term positive shift over the late Miocene (Figure 7e), which may indicate a cooling of deep water and/or intensified Antarctic glaciation. A positive shift in seawater $\delta^{18}O$ by per 0.1‰ corresponds to ~0.5°C cooling, or ~6.25 m ice-volume equivalent sealevel drop (Miller et al., 2020). However, the general pattern of benthic $\delta^{18}O$ in Hole U1560A differs from that of CENOGRID (Westerhold et al., 2020) (Figure 7e) and the composited benthic $\delta^{18}O$ reported by Zachos et al. (2001). These three benthic $\delta^{18}O$ curves differ from each other, as the $\delta^{18}O$ originate in different sites in global oceans (Figure S10 in Supporting Information S1). This suggests that the late Miocene deep ocean $\delta^{18}O$ can be rather influenced by regional hydrological or circulation conditions, since the high-frequency $\delta^{18}O$ dynamic is relatively small, that is, ~0.5‰. Hole U1560A (30°S, 3,723 m) in the South Atlantic basin is bathed in the AABW (Figure S11 in Supporting Information S1) and close to the location, that is, the Weddell Sea, where the deep water is formed in the Atlantic sector. Comparatively, the CENOGRID $\delta^{18}O$ are adjusted to Site U1337 in the equatorial Eastern Pacific (3.8°N, 4,463 m) (Westerhold et al., 2020), which may include the signal of northern-sourced deep water. It can be inferred that the benthic $\delta^{18}O$ in Hole U1560A may rather reflect the surface water conditions where the AABW is formed. Therefore, the step-wise positive shift of benthic $\delta^{18}O$ in Hole U1560A can indicate either cooling of the surface water in the Southern Ocean or intensified Antarctic glaciation, which facilitates a sluggish or weakening of overturning circulation.

The major benthic $\delta^{18}O$ positive shift by ~0.2‰ is observed at 7.7–8.2 Ma in Hole U1560A (Figure 7e), which coincides with the interval when the basinal $\delta^{13}C$ gradient between the Atlantic and Pacific Oceans becomes greater (Figure 7d). The intensified basinal $\delta^{13}C$ gradient at ~7 Ma also agrees with a relatively weaker positive shift in benthic $\delta^{18}O$ by ~0.1‰ in Hole U1560A, as well as coincides with a positive shift by ~0.2‰ in the CENOGRID $\delta^{18}O$ (Figure 7e). Therefore, based on the lines of evidence from the box-modeling and observations, we can propose that the increased basinal $\delta^{13}C$ gradient was caused by sluggish or weakening of deep ocean meridional overturning circulation due to enhanced Antarctic glaciation.

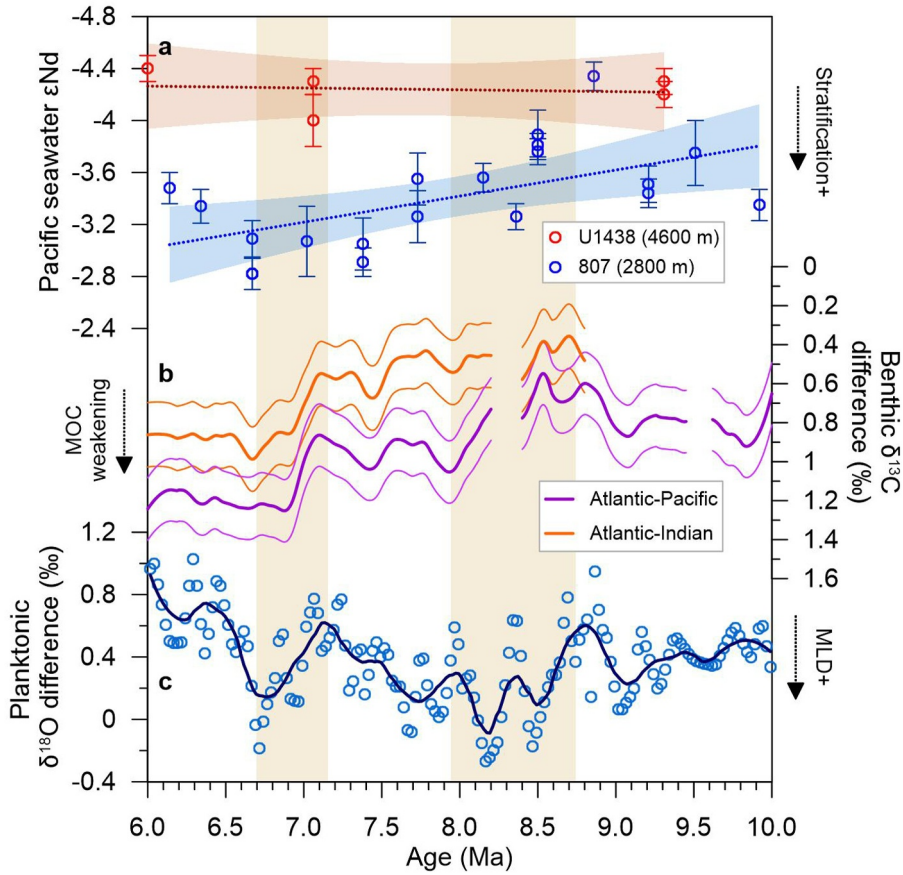


Figure 9. Comparison of the paleoceanographic records over the late Miocene. (a) The Pacific seawater neodymium isotope (εNd, $\pm1\sigma$) recorded at Site 807 (2,800 m depth) (Le Houedec et al., 2016) and Site U1438 (4,600 m) (Kender et al., 2018), with their linear fits and 95% intervals. Greater εNd differences indicate increased stratification between the intermediate and deep layers; (b) Deep ocean $\delta^{13}C$ gradient ($\pm1\sigma$) between the Atlantic and Indian, Pacific Oceans. Increased gradient suggests weakened meridional overturning circulation (MOC); (c) The planktonic foraminiferal $\delta^{18}O$ differences between the subsurface and surface species at Site 806 in the tropical Western Pacific (LaRiviere et al., 2012). Smaller differences indicate deeper mixed layer depth (MLD). The bold curve indicates 11-point running average. Shadings indicate the intervals with weakened MOC.

We note that the present box-model cannot simulate the deep ocean organic matter (OM) transport and degradation along the ocean conveyor belt. Assuming the OM degradation follows a first-order kinetic, a simple one-dimensional model of deep ocean $\delta^{13}C$ is given by

$$\delta_c = \frac{[DIC]_0 * (\delta_c)_0 + [OM]_0 * \left(1 - e^{-k\tau_L^l}\right) * \left((\delta_c)_0 + \epsilon_p\right)}{[DIC]_0 + [OM]_0 * \left(1 - e^{-k\tau_L^l}\right)}, \quad (3)$$

where k (s^{-1}) is the rate constant of OM degradation, τ (s) is the residence time of OM in deep water, $\frac{l}{L}$ represents the relative distance of OM transport. This simple equation can also explain that sluggish of deep ocean circulation with larger τ will cause increased basinal $\delta^{13}C$ gradient.

3.3.3. Changes in Riverine Carbon $\delta^{13}C$

Finally, the steady state condition was perturbed by reducing riverine carbon $\delta^{13}C$ input to ocean. Du et al. (2022) propose that the riverine carbon $\delta^{13}C$ should decrease, since more ^{13}C is sequestered in land as a result of C_4-plant expansion. This process can be a driver for the LMCIS. Here an experiment is designed to evaluate the its

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influence on the global deep ocean $\delta^{13}C$ pattern. We assume that 70% of the riverine carbon was input to Indian-Pacific Box and 30% to Atlantic.

After the perturbation, the system takes $\sim 10^5$–10^6 years to arrive at steady state. Such a long recovery time should result from the small riverine carbon flux relative to the large ocean carbon reservoir, indicating a long residence time of the land-ocean carbon cycling in an order of magnitude of $\sim 10^5$ years. We show that by reducing the riverine carbon $\delta^{13}C$ by 1‰, there is a long-term decline in $\delta^{13}C$ by 1‰ in all boxes, but with invariant gradient between the deep Atlantic and Indian-Pacific boxes (Figure 8f). The present modeling results support the reorganization of land ecosystem as a driver for the LMCIS, however, reducing the riverine carbon $\delta^{13}C$ would not change the basinal $\delta^{13}C$ gradient. Additionally, if the ratio of riverine carbon input to Indian-Pacific Box was reduced from 70% to 30%, the box-modeling shows that $\Delta\delta^{13}C_{Atl.-Pac.}$ would decrease by 0.04‰. This indicates that the timing of land ecosystem evolution in different lands, for example, Asia, Africa, and America, which governs the riverine carbon input to surrounding oceans, may potentially influence the global deep ocean $\delta^{13}C$ pattern.

Besides through riverine carbon input, atmospheric CO_2 cycling is supposed to be another way for converting the signals of land carbon cycle perturbation to ocean. For example, the degree of carbon isotopic fractionation through photosynthesis by C_4-plants should be weaker than C_3-plants, as C_4-plants have more positive $\delta^{13}C$ values. The C_4-grassland expansion would lead to a more negative atmospheric CO_2 $\delta^{13}C$ signature which can be transported to ocean through air-sea CO_2 exchange. However, our box-model does not simulate these processes, since the total land carbon quantity, the ratio of C_3/C_4 carbon, and the land-based net carbon fixing rate are poorly known.

3.4. Implications for the Late Miocene Climate Change

We suggest the sluggish or weakening of ocean overturning circulation as a driver for the intensified basinal $\delta^{13}C$ gradient over the late Miocene, and that enhanced Antarctic glaciation can be responsible for such an ocean circulation change. Over the late Miocene, the climate system and its interaction with carbon cycle have changed from a dominant low-latitude control to both low- and high-latitude (De Vleeschouwer et al., 2020; X. Ma et al., 2022). Before the late Miocene cooling or glaciation, glacial periods (high ocean $\delta^{18}O$) occur at eccentricity minima with weaker low-latitude hydrological cycles. A weaker chemical weathering can be inferred, leading to less terrestrial carbon input to ocean (high ocean $\delta^{13}C$). As such, the benthic $\delta^{18}O$ and $\delta^{13}C$ are presenting as in-phase. After the late Miocene with enhanced high-latitude glaciation, stronger glacial-interglaical transition present a quasi-periodicity averaging 100-kyr duration, for example, during the late Pleistocene. Intensified ice-sheet covering (high ocean $\delta^{18}O$) would cause a contraction of land ecosystem in high-latitudes, and more terrestrial carbon was input to ocean, lowering oceanic $\delta^{13}C$. As such, the benthic $\delta^{18}O$ and $\delta^{13}C$ are presenting in anti-phase (De Vleeschouwer et al., 2020; X. Ma et al., 2022). Therefore, the late Miocene is a critical period, as the interactions between carbon cycle and Earth's climate have significantly changed.

There is a general agreement that atmospheric CO_2, as a kind of greenhouse gas, is a major driver for Earth's climate during the Cenozoic (CenCO2PIP-Consortium, 2023). For an example of the Neogene period, the Global Mean Surface Temperature (GMST) are here plotted versus the atmospheric CO_2 concentration during the Mid-Miocene Climate Transition (MMCT) cooling around 13.9 Ma (Hansen et al., 2023; Sosdian et al., 2018), the Pliocene-Pleistocene cooling during 2.3–3.3 Ma (Clark et al., 2024; Martínez-Botí et al., 2015), and the Pleistocene glacial-interglacial cycles over the last 800 kyr (Lüthi et al., 2008) (Figure 10). Although with different Earth System Sensitivity (ESS, the GMST change with respect to CO_2 doubling), the paleo-climate reconstructions/records show a CO_2 radiative forcing on the GMST during these critical climate transitions (Figure 10). However, the late Miocene period is mysterious. The late Miocene cooling is supposed to lead a decline in GMST by ~1–2°C between the MMCT and Pliocene (Clark et al., 2024; Hansen et al., 2023), while the atmospheric CO_2 concentrations were comparable during these periods (Figure 10). The decoupling of CO_2 and GMST suggests non-CO_2 forcing operated in the climate system during the late Miocene.

The ocean carbon cycle and circulation changes may provide insights into the late Miocene climate change. In a first-order, the late Miocene expansion of C_4-grassland together with a shrinkage of C_3-forests may have contributed to an increase in Earth surface albedo. Increased planetary albedo would cause more solar short-

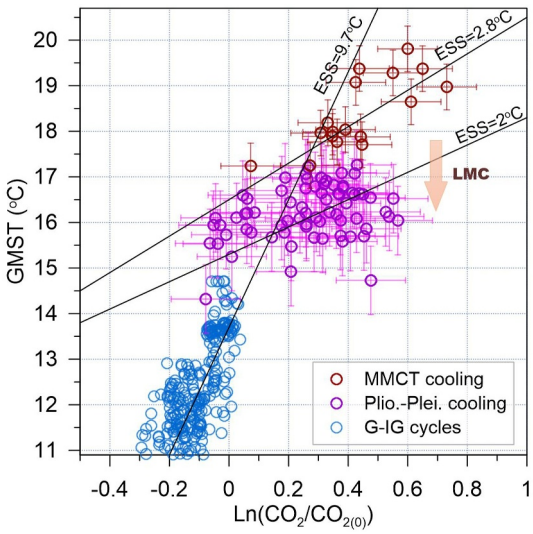


Figure 10. The Global Mean Surface Temperature (GMST) co-varied with CO_2 during the critical climate transitions since the Neogene (since 23 Ma). For example, the Mid-Miocene Climate Transition (MMCT) cooling (across 13.9 Ma), the Pliocene-Pleistocene (Plio.-Plei.) cooling (2.3–3.3 Ma), and the last 800 kyr glacial-interglacial cycles are coupling with CO_2, suggesting a dominant CO_2 radiative forcing. The data show that the GMST difference between the MMCT and Pliocene may be ~1–2°C, revealing the late Miocene cooling (LMC). However, atmospheric CO_2 concentrations during these two periods were comparable, suggesting a non-CO_2 forcing. The GMST over the last 5 Myr (since Pliocene) refer to Clark et al. (2024), and the Miocene GMST refer to Hansen et al. (2023). The CO_2 data across the MMCT, the Plio.-Plei. cooling, and during the last 800 kyr are from Sosdian et al. (2018) (boron), Martínez-Botí et al. (2015) (boron), and Lüthi et al. (2008) (ice-core), respectively. All the error bars here indicate their 1σ uncertainties. The CO_2 forcing is expressed by $Ln(CO_2/CO_{2(0)})$, where $CO_{2(0)}$ indicates the preindustrial level of 278 ppm. The lines are slopes of changing GMST with respect to CO_2 forcing, indicating the Earth System Sensitivity (ECS, or temperature rise with CO_2 doubling).

radiative energy reflection to space, lowering the GMST. Köhler et al. (2010) estimated that such changes in land (vegetation and cryosphere)-based albedo will cause a temperature variation of ~1.3°C over the late Pleistocene glacial-interglaical cycles. This kind of non-CO_2 forcing mechanism is also validated by a general circulation model (Knorr et al., 2011). The authors show that weaker Earth surface albedo due to the Miocene vegetation types can explain ~2.5°C higher GMST than the modern, with comparable atmospheric CO_2 concentration (~350 ppm) (Knorr et al., 2011). The late Miocene cooling may link to or facilitate enhanced Antarctic glaciation, which would in turn reorganize the global thermohaline circulation. The meridional overturning circulation weakening as revealed by the intensified basinal $\delta^{13}C$ gradient may be accompanied by a reduction of the polarward ocean heat transport (Winton et al., 2013; X. Liu et al., 2024). As a result, more heat would be stored in the tropical oceans, which can be indicated by the step-wisely reduced surface and subsurface planktonic foraminiferal $\delta^{18}O$ difference recorded in the tropical Western Pacific (Site 806) (LaRiviere et al., 2012) during the late Miocene (Figure 9c). The decreased $\delta^{18}O$ differences, which indicate mixed layer (or thermocline) deepening, generally coincide with the increased basinal $\delta^{13}C$ gradient during 8–8.5 Ma and 6.9–7.2 Ma (Figures 9b and 9c), suggesting the control of ocean circulation on global ocean heat distribution. Notably, a long-term increase in the surface and subsurface planktonic foraminiferal $\delta^{18}O$ difference is also observed (Figure 9c), which can result from the development of "cold-tongue" in the equatorial Eastern Pacific due to the close of Central American Seaway during the late Miocene (J. Liu et al., 2019).

4. Summary

This study presents new late Miocene benthic foraminiferal isotopes in Hole U1560A in the South Atlantic. The down-core sediments were retrieved during IODP Expeditions 390 & 393: "South Atlantic Transect." Together with existing benthic foraminiferal $\delta^{13}C$, we have examined the evolution of global deep ocean $\delta^{13}C$ pattern over the late Miocene. Our analysis shows a step-wise increase in deep ocean $\delta^{13}C$ difference between the Atlantic and Indian-Pacific Oceans over the late Miocene, suggesting intensified basinal $\delta^{13}C$ gradient. As supported by lines of evidence from geological observations and a diagnostic carbon cycle box-model developed in this study, we suggest that the intensified basinal $\delta^{13}C$ can result from sluggish or weakening of ocean meridional overturning circulation triggered by enhanced Antarctic glaciation. The weakening of overturning circulation may play a role in Earth's climate system by reducing the ocean heat transport to high-latitudes.

Appendix A: Expeditions 390 & 393 Scientists' Affiliations

EXP. 390 scientists: Rosalind M. Coggon[1], Jason B. Sylvan[2], Emily R. Estes[3], Masataka Aizawa[4], Chiara Borrelli[5], Joshua D. Bridges[5], Elliot J. Carter[6], Nobuhiro Doi[7], Justin D. Estep[8], William P. Gilhooly III[9], Lewis Grant[1], Michael R. Kaplan[10], Pamela D. Kempton[11], Walter Kurz[12], Christopher M. Lowery[13], Andrew McIntyre[14], Muthusamy Prakasam[15], Claire M. Routledge[16], Angela L. Slagle[10], Mako Takada[17], Leonardo Tamborrino[18], Liyan Tian[19], Yi Wang[20], Kiho Yang[21], Tiantian Yu[22], Jaume Dinarès-Turell[23].

EXP. 393 scientists: Damon A.H. Teagle[1], Julia Reece[24], Trevor J. Williams[3], Elmar Albers[25], Chiara Amadori[26], Thomas M. Belgrano[1], Timothy D'Angelo[27], Aled Evans[1], Gilles M. Guerin[10], Michelle Harris[28], Victoria M. Hojnacki[29], Gilbert Hong[30], Mallika Jonnalagadda[31], Daisuke Kuwano[32], Jessica Labonte[33], Adriane R. Lam[34], Marcin Latas[35], Wanyi Lu[36,37], Paul Moal-Darrigade[38], Stephen F. Pekar[39], Claudio Robustelli Test[40], Jeffrey G.

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Data Availability Statement

The late Miocene benthic foraminiferal isotopes and age model in Hole U1560A, the calculated benthic foraminiferal $\delta^{13}C$ differences, and the Matlab codes for the carbon cycle box-model are available in Jin (2025).

References

Berger, W. (1982). Increase of carbon dioxide in the atmosphere during deglaciation: The coral reef hypothesis. *Naturwissenschaften*, *69*(2), 87–88. https://doi.org/10.1007/bf00441228

Bickert, T., Haug, G. H., & Tiedemann, R. (2004). Late Neogene benthic stable isotope record of ocean drilling program site 999: Implications for Caribbean paleoceanography, organic carbon burial, and the messinian salinity crisis. *Paleoceanography*, *19*(1), PA1023. https://doi.org/10.1029/2002pa000799

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Billups, K. (2002). Late Miocene through early Pliocene deep water circulation and climate change viewed from the sub-Antarctic south Atlantic. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *185*(3–4), 287–307. https://doi.org/10.1016/s0031-0182(02)00340-1

Bolton, C. T., Gray, E., Kuhnt, W., Holbourn, A. E., Lübbers, J., Grant, K., et al. (2022). Secular and orbital-scale variability of equatorial Indian Ocean summer monsoon winds during the late Miocene. *Climate of the Past*, *18*(4), 713–738. https://doi.org/10.5194/cp-18-713-2022

Bolton, C. T., Hernández-Sánchez, M. T., Fuertes, M.-Á., González-Lemos, S., Abrevaya, L., Mendez-Vicente, A., et al. (2016). Decrease in coccolithophore calcification and CO_2 since the middle Miocene. *Nature Communications*, *7*(1), 10284. https://doi.org/10.1038/ncomms10284

Brown, R., Chalk, T., Crocker, A., Wilson, P., & Foster, G. (2022). Late Miocene cooling coupled to carbon dioxide with Pleistocene-like climate sensitivity. *Nature Geoscience*, *15*(8), 664–670. https://doi.org/10.1038/s41561-022-00982-7

Cerling, T. E., Wang, Y., & Quade, J. (1993). Expansion of C_4 ecosystems as an indicator of global ecological change in the late Miocene. *Nature*, *361*(6410), 344–345. https://doi.org/10.1038/361344a0

Clark, P. U., Shakun, J. D., Rosenthal, Y., Köhler, P., & Bartlein, P. J. (2024). Global and regional temperature change over the past 4.5 million years. *Science*, *383*(6685), 884–890. https://doi.org/10.1126/science.adi1908

Coggon, R., Teagle, D., Sylvan, J., Reece, J., Estes, E., Williams, T., et al. (2024). In *Proceedings of the international ocean discovery program, 390/393*. College station, tx. https://doi.org/10.14379/iodp.proc.390393.2024

Cramer, B., Toggweiler, J., Wright, J., Katz, M., & Miller, K. (2009). Ocean overturning since the late cretaceous: Inferences from a new benthic foraminiferal isotope compilation. *Paleoceanography*, *24*(4). https://doi.org/10.1029/2008pa001683

Cui, Y., Schubert, B. A., & Jahren, A. H. (2020). A 23 my record of low atmospheric CO_2. *Geology*, *48*(9), 888–892. https://doi.org/10.1130/g47681.1

De Vleeschouwer, D., Drury, A. J., Vahlenkamp, M., Rochholz, F., Liebrand, D., & Pälike, H. (2020). High-latitude biomes and rock weathering mediate climate–carbon cycle feedbacks on eccentricity timescales. *Nature Communications*, *11*(1), 5013. https://doi.org/10.1038/s41467-020-18733-w

Diester-Haass, L., Billups, K., & Emeis, K. C. (2005). In search of the late Miocene–early Pliocene "biogenic bloom" in the Atlantic Ocean (ocean drilling program sites 982, 925, and 1088). *Paleoceanography*, *20*(4). https://doi.org/10.1029/2005pa001139

Diester-Haass, L., Billups, K., & Emeis, K. C. (2006). Late Miocene carbon isotope records and marine biological productivity: Was there a (dusty) link? *Paleoceanography*, *21*(4), PA4216. https://doi.org/10.1029/2006pa001267

Drury, A. J., Lee, G., Gray, W. R., Lyle, M., Westerhold, T., Shevenell, A. E., & John, C. M. (2018). Deciphering the state of the late Miocene to early Pliocene equatorial Pacific. *Paleoceanography and Paleoclimatology*, *33*(3), 246–263. https://doi.org/10.1002/2017pa003245

Drury, A. J., Westerhold, T., Frederichs, T., Tian, J., Wilkens, R., Channell, J. E., et al. (2017). Late Miocene climate and time scale reconciliation: Accurate orbital calibration from a deep-sea perspective. *Earth and Planetary Science Letters*, *475*, 254–266. https://doi.org/10.1016/j.epsl.2017.07.038

Drury, A. J., Westerhold, T., Hodell, D., & Röhl, U. (2018). Reinforcing the north Atlantic backbone: Revision and extension of the composite splice at ODP Site 982. *Climate of the Past*, *14*(3), 321–338. https://doi.org/10.5194/cp-14-321-2018

Du, J., Tian, J., & Ma, W. (2022). The late Miocene carbon isotope shift driven by synergetic terrestrial processes: A box-model study. *Earth and Planetary Science Letters*, *584*, 117457. https://doi.org/10.1016/j.epsl.2022.117457

Ferrari, R., Jansen, M. F., Adkins, J. F., Burke, A., Stewart, A. L., & Thompson, A. F. (2014). Antarctic sea ice control on ocean circulation in present and glacial climates. *Proceedings of the National Academy of Sciences of the United States of America*, *111*(24), 8753–8758. https://doi.org/10.1073/pnas.1323922111

Freeman, K., & Colarusso, L. (2001). Molecular and isotopic records of C_4 grassland expansion in the late Miocene. *Geochimica et Cosmochimica Acta*, *65*(9), 1439–1454. https://doi.org/10.1016/s0016-7037(00)00573-1

Gradstein, F. M., & Ogg, J. G. (2020). The chronostratigraphic scale. In *Geologic time scale 2020* (pp. 21–32). Elsevier.

Gulick, S. P., Shevenell, A. E., Montelli, A., Fernandez, R., Smith, C., Warny, S., et al. (2017). Initiation and long-term instability of the East Antarctic ice sheet. *Nature*, *552*(7684), 225–229. https://doi.org/10.1038/nature25026

Hansen, J. E., Sato, M., Simons, L., Nazarenko, L. S., Sangha, I., Kharecha, P., et al. (2023). Global warming in the pipeline. *Oxford Open Climate Change*, *3*(1), kgad008. https://doi.org/10.1093/oxfclm/kgad008

Herbert, T. D., Lawrence, K. T., Tzanova, A., Peterson, L. C., Caballero-Gill, R., & Kelly, C. S. (2016). Late Miocene global cooling and the rise of modern ecosystems. *Nature Geoscience*, *9*(11), 843–847. https://doi.org/10.1038/ngeo2813

Hilgen, F. (2025). Revised astrochronology for the lower to middle Miocene of IODP Site u1337. Implications for the Cenozoic megasplice (CENOGRID). *Palaeogeography, Palaeoclimatology, Palaeoecology*, *662*, 112748. https://doi.org/10.1016/j.palaeo.2025.112748

Hodell, D. A., Curtis, J. H., Sierro, F. J., & Raymo, M. E. (2001). Correlation of late Miocene to early Pliocene sequences between the Mediterranean and North Atlantic. *Paleoceanography*, *16*(2), 164–178. https://doi.org/10.1029/1999pa000487

Hodell, D. A., & Venz-Curtis, K. A. (2006). Late Neogene history of deepwater ventilation in the southern ocean. *Geochemistry, Geophysics, Geosystems*, *7*(9), Q09001. https://doi.org/10.1029/2005gc001211

Holbourn, A., Kuhnt, W., Frank, M., & Haley, B. A. (2013). Changes in Pacific Ocean circulation following the Miocene onset of permanent Antarctic ice cover. *Earth and Planetary Science Letters*, *365*, 38–50. https://doi.org/10.1016/j.epsl.2013.01.020

Holbourn, A., Kuhnt, W., Kulhanek, D. K., Mountain, G., Rosenthal, Y., Sagawa, T., et al. (2024). Re-organization of Pacific overturning circulation across the Miocene climate optimum. *Nature Communications*, *15*(1), 8135. https://doi.org/10.1038/s41467-024-52516-x

Holbourn, A., Kuhnt, W., Schulz, M., Flores, J.-A., & Andersen, N. (2007). Orbitally-paced climate evolution during the middle Miocene "monterey" carbon-isotope excursion. *Earth and Planetary Science Letters*, *261*(3–4), 534–550. https://doi.org/10.1016/j.epsl.2007.07.026

Holbourn, A. E., Kuhnt, W., Clemens, S. C., Kochhann, K. G., Jöhnck, J., Lübbers, J., & Andersen, N. (2018). Late Miocene climate cooling and intensification of Southeast Asian winter monsoon. *Nature Communications*, *9*(1), 1584. https://doi.org/10.1038/s41467-018-03950-1

Hönisch, B., Royer, D. L., Breecker, D. O., Polissar, P. J., Bowen, G. J., Henehan, M. J., et al. (2023). Toward a Cenozoic history of atmospheric CO_2. *Science*, *382*(6675), eadi5177. https://doi.org/10.1126/science.adi5177

Hüsing, S., Hilgen, F., Aziz, H. A., & Krijgsman, W. (2007). Completing the Neogene geological time scale between 8.5 and 12.5 ma. *Earth and Planetary Science Letters*, *253*(3–4), 340–358. https://doi.org/10.1016/j.epsl.2006.10.036

Hüsing, S., Kuiper, K., Link, W., Hilgen, F. J., & Krijgsman, W. (2009). The upper Tortonian–lower Messinian at Monte dei Corvi (northern Apennines, Italy): Completing a Mediterranean reference section for the Tortonian stage. *Earth and Planetary Science Letters*, *282*(1–4), 140–157. https://doi.org/10.1016/j.epsl.2009.03.010

Ji, S., Nie, J., Lechler, A., Huntington, K. W., Heitmann, E. O., & Breecker, D. O. (2018). A symmetrical CO_2 peak and asymmetrical climate change during the middle Miocene. *Earth and Planetary Science Letters*, *499*, 134–144. https://doi.org/10.1016/j.epsl.2018.07.011

Jin, X. (2025). *[collection] sat results: Benthic isotopes*. Figshare. https://doi.org/10.6084/m9.figshare.28120535

Karp, A. T., Behrensmeyer, A. K., & Freeman, K. H. (2018). Grassland fire ecology has roots in the late Miocene. *Proceedings of the National Academy of Sciences of the United States of America*, *115*(48), 12130–12135. https://doi.org/10.1073/pnas.1809758115

25724525, 2025, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024PA005028 by Tongji University, Wiley Online Library on [30/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Katz, M. E., Wright, J. D., Miller, K. G., Cramer, B. S., Fennel, K., & Falkowski, P. G. (2005). Biological overprint of the geological carbon cycle. *Marine Geology*, *217*(3–4), 323–338. https://doi.org/10.1016/j.margeo.2004.08.005

Kender, S., Bogus, K. A., Cobb, T. D., & Thomas, D. J. (2018). Neodymium evidence for increased circumpolar deep water flow to the North Pacific during the middle Miocene climate transition. *Paleoceanography and Paleoclimatology*, *33*(7), 672–682. https://doi.org/10.1029/2017pa003309

Knorr, G., Butzin, M., Micheels, A., & Lohmann, G. (2011). A warm Miocene climate at low atmospheric CO_2 levels. *Geophysical Research Letters*, *38*(20), L20701. https://doi.org/10.1029/2011gl048873

Köhler, P., Bintanja, R., Fischer, H., Joos, F., Knutti, R., Lohmann, G., & Masson-Delmotte, V. (2010). What caused Earth's temperature variations during the last 800,000 years? Data-based evidence on radiative forcing and constraints on climate sensitivity. *Quaternary Science Reviews*, *29*(1–2), 129–145. https://doi.org/10.1016/j.quascirev.2009.09.026

Kump, L. R., & Arthur, M. A. (1999). Interpreting carbon-isotope excursions: Carbonates and organic matter. *Chemical Geology*, *161*(1–3), 181–198. https://doi.org/10.1016/s0009-2541(99)00086-8

LaRiviere, J. P., Ravelo, A. C., Crimmins, A., Dekens, P. S., Ford, H. L., Lyle, M., & Wara, M. W. (2012). Late Miocene decoupling of oceanic warmth and atmospheric carbon dioxide forcing. *Nature*, *486*(7401), 97–100. https://doi.org/10.1038/nature11200

Laskar, J., Robutel, P., Joutel, F., Gastineau, M., Correia, A. C., & Levrard, B. (2004). A long-term numerical solution for the insolation quantities of the Earth. *Astronomy & Astrophysics*, *428*(1), 261–285. https://doi.org/10.1051/0004-6361:20041335

Lauvset, S. K., Lange, N., Tanhua, T., Bittig, H. C., Olsen, A., Kozyr, A., et al. (2023). Global ocean data analysis project version 2.2023 (glodapv2. 2023).

Le Houedec, S., Meynadier, L., & Allègre, C. J. (2016). Seawater Nd isotope variation in the Western Pacific ocean since 80 ma (ODP 807, ontong java plateau). *Marine Geology*, *380*, 138–147. https://doi.org/10.1016/j.margeo.2016.07.005

Li, M., Hinnov, L., & Kump, L. (2019). Acycle: Time-series analysis software for paleoclimate research and education. *Computers & Geosciences*, *127*, 12–22. https://doi.org/10.1016/j.cageo.2019.02.011

Liu, F., Du, J., Huang, E., Ma, W., Ma, X., Lourens, L. J., & Tian, J. (2024). Accelerated marine carbon cycling forced by tectonic degassing over the Miocene climate optimum. *Science Bulletin*, *69*(6), 823–832. https://doi.org/10.1016/j.scib.2023.12.052

Liu, J., Tian, J., Liu, Z., Herbert, T. D., Fedorov, A. V., & Lyle, M. (2019). Eastern equatorial Pacific cold tongue evolution since the late Miocene linked to extratropical climate. *Science Advances*, *5*(4), eaau6060. https://doi.org/10.1126/sciadv.aau6060

Liu, X., Herold, N., & Huber, M. (2024). Atlantic meridional overturning circulation influence on the annual mean intertropical convergence zone location in the Miocene. *Geophysical Research Letters*, *51*(9), e2024GL109159. https://doi.org/10.1029/2024gl109159

Lüthi, D., Le Floch, M., Bereiter, B., Blunier, T., Barnola, J.-M., Siegenthaler, U., et al. (2008). High-resolution carbon dioxide concentration record 650,000–800,000 years before present. *Nature*, *453*(7193), 379–382. https://doi.org/10.1038/nature06949

Ma, W., Tian, J., Li, Q., & Wang, P. (2011). Simulation of long eccentricity (400-kyr) cycle in ocean carbon reservoir during Miocene climate optimum: Weathering and nutrient response to orbital change. *Geophysical Research Letters*, *38*(10). https://doi.org/10.1029/2011gl047680

Ma, X., Ma, W., Tian, J., Yu, J., & Huang, E. (2022). Ice sheet and terrestrial input impacts on the 100-kyr ocean carbon cycle during the middle Miocene. *Global and Planetary Change*, *208*, 103723. https://doi.org/10.1016/j.gloplacha.2021.103723

Martínez-Botí, M. A., Foster, G. L., Chalk, T. B., Rohling, E. J., Sexton, P. F., Lunt, D. J., et al. (2015). Plio-Pleistocene climate sensitivity evaluated using high-resolution CO_2 records. *Nature*, *518*(7537), 49–54. https://doi.org/10.1038/nature14145

Miao, Y., Herrmann, M., Wu, F., Yan, X., & Yang, S. (2012). What controlled mid–late Miocene long-term aridification in central Asia? Global cooling or Tibetan Plateau uplift: A review. *Earth-Science Reviews*, *112*(3–4), 155–172. https://doi.org/10.1016/j.earscirev.2012.02.003

Miller, K. G., Browning, J. V., Schmelz, W. J., Kopp, R. E., Mountain, G. S., & Wright, J. D. (2020). Cenozoic sea-level and cryospheric evolution from deep-sea geochemical and continental margin records. *Science Advances*, *6*(20), eaaz1346. https://doi.org/10.1126/sciadv.aaz1346

Ogg, J. G. (2020). Geomagnetic polarity time scale. In *Geologic time scale 2020* (pp. 159–192). Elsevier.

Palike, H., Norris, R. D., Herrle, J. O., Wilson, P. A., Coxall, H. K., Lear, C. H., et al. (2006). The heartbeat of the Oligocene climate system. *Science*, *314*(5807), 1894–1898. https://doi.org/10.1126/science.1133822

Polissar, P. J., Uno, K. T., Phelps, S. R., Karp, A. T., Freeman, K. H., & Pensky, J. L. (2021). Hydrologic changes drove the late Miocene expansion of C_4 grasslands on the northern Indian subcontinent. *Paleoceanography and Paleoclimatology*, *36*(4), e2020PA004108. https://doi.org/10.1029/2020pa004108

Raffi, I., Wade, B., Pälike, H., Beu, A., Cooper, R., Crundwell, M., et al. (2020). The Neogene period. In *Geologic time scale 2020* (pp. 1141–1215). Elsevier.

Schlitzer, R. (2024). Ocean data view. Retrieved from https://odv.awi.de

Shen, X., Wan, S., Colin, C., Tada, R., Shi, X., Pei, W., et al. (2018). Increased seasonality and aridity drove the C_4 plant expansion in central Asia since the Miocene–Pliocene boundary. *Earth and Planetary Science Letters*, *502*, 74–83. https://doi.org/10.1016/j.epsl.2018.08.056

Skinner, L., McCave, I., Carter, L., Fallon, S., Scrivner, A., & Primeau, F. (2015). Reduced ventilation and enhanced magnitude of the deep Pacific carbon pool during the last glacial period. *Earth and Planetary Science Letters*, *411*, 45–52. https://doi.org/10.1016/j.epsl.2014.11.024

Sosdian, S. M., Greenop, R., Hain, M., Foster, G. L., Pearson, P. N., & Lear, C. H. (2018). Constraining the evolution of Neogene ocean carbonate chemistry using the boron isotope PH proxy. *Earth and Planetary Science Letters*, *498*, 362–376. https://doi.org/10.1016/j.epsl.2018.06.017

Super, J. R., Thomas, E., Pagani, M., Huber, M., O'Brien, C., & Hull, P. M. (2018). North Atlantic temperature and pCO_2 coupling in the early-middle Miocene. *Geology*, *46*(6), 519–522. https://doi.org/10.1130/g40228.1

Talley, L. D. (2013). Closure of the global overturning circulation through the Indian, Pacific, and southern oceans: Schematics and transports. *Oceanography*, *26*(1), 80–97. https://doi.org/10.5670/oceanog.2013.07

Tanner, T., Hernández-Almeida, I., Drury, A. J., Guitián, J., & Stoll, H. (2020). Decreasing atmospheric CO_2 during the late Miocene cooling. *Paleoceanography and Paleoclimatology*, *35*(12), e2020PA003925. https://doi.org/10.1029/2020pa003925

Tian, J., Ma, X., Zhou, J., Jiang, X., Lyle, M., Shackford, J., & Wilkens, R. (2018). Paleoceanography of the east equatorial Pacific over the past 16 myr and Pacific–Atlantic comparison: High resolution benthic foraminiferal $\delta^{18}O$ and $\delta^{13}C$ records at IODP Site U1337. *Earth and Planetary Science Letters*, *499*, 185–196. https://doi.org/10.1016/j.epsl.2018.07.025

Uno, K. T., Polissar, P. J., Jackson, K. E., & de Menocal, P. B. (2016). Neogene biomarker record of vegetation change in eastern Africa. *Proceedings of the National Academy of Sciences of the United States of America*, *113*(23), 6355–6363. https://doi.org/10.1073/pnas.1521267113

Wang, X. T., Wang, Y., Auderset, A., Sigman, D. M., Ren, H., Martínez-García, A., et al. (2022). Oceanic nutrient rise and the late Miocene inception of Pacific oxygen-deficient zones. *Proceedings of the National Academy of Sciences of the United States of America*, *119*(45), e2204986119. https://doi.org/10.1073/pnas.2204986119

25724525, 2025, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024PA005028 by Tongji University, Wiley Online Library on [30/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Wen, Y., Zhang, L., Holbourn, A. E., Zhu, C., Huntington, K. W., Jin, T., et al. (2023). CO_2-forced late Miocene cooling and ecosystem reorganizations in East Asia. *Proceedings of the National Academy of Sciences of the United States of America*, *120*(5), e2214655120. https://doi.org/10.1073/pnas.2214655120

Westerhold, T., Agnini, C., Anagnostou, E., Hilgen, F., Hönisch, B., Meckler, A., et al. (2024). Timing is everything. *Paleoceanography and Paleoclimatology*, *39*(12), e2024PA004932. https://doi.org/10.1029/2024pa004932

Westerhold, T., Marwan, N., Drury, A. J., Liebrand, D., Agnini, C., Anagnostou, E., et al. (2020). An astronomically dated record of Earth's climate and its predictability over the last 66 million years. *Science*, *369*(6509), 1383–1387. https://doi.org/10.1126/science.aba6853

Williams, T., van de Flierdt, T., Hemming, S. R., Chung, E., Roy, M., & Goldstein, S. L. (2010). Evidence for iceberg armadas from east Antarctica in the southern ocean during the late Miocene and early Pliocene. *Earth and Planetary Science Letters*, *290*(3–4), 351–361. https://doi.org/10.1016/j.epsl.2009.12.031

Winton, M., Griffies, S. M., Samuels, B. L., Sarmiento, J. L., & Frölicher, T. L. (2013). Connecting changing ocean circulation with changing climate. *Journal of Climate*, *26*(7), 2268–2278. https://doi.org/10.1175/jcli-d-12-00296.1

Wright, J. D., & Miller, K. G. (1996). Control of North Atlantic deep water circulation by the Greenland-Scotland ridge. *Paleoceanography*, *11*(2), 157–170. https://doi.org/10.1029/95pa03696

Zachos, J., Pagani, M., Sloan, L., Thomas, E., & Billups, K. (2001). Trends, rhythms, and aberrations in global climate 65 ma to present. *Science*, *292*(5517), 686–693. https://doi.org/10.1126/science.1059412

Zhang, Q., Zhang, R., Hao, Q., Clift, P. D., Roberts, A. P., Florindo, F., et al. (2024). East Asian winter monsoon intensification over the northwest Pacific Ocean driven by late Miocene atmospheric CO_2 decline. *Science Advances*, *10*(25), eadm8270. https://doi.org/10.1126/sciadv.adm8270

Zhang, Y. G., Pagani, M., Henderiks, J., & Ren, H. (2017). A long history of equatorial deep-water upwelling in the Pacific Ocean. *Earth and Planetary Science Letters*, *467*, 1–9. https://doi.org/10.1016/j.epsl.2017.03.016

25724525, 2025, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024PA005028 by Tongji University, Wiley Online Library on [30/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License